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Deep polar transformer semantic coding framework for reliable 6G autonomous vehicle communication

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ABSTRACT

The development of sixth generation (6G) networks for Autonomous Vehicles (AVs) requires an unprecedented level of ultra-reliable low-latency communication (URLLC). However, existing communication paradigms such as separate source channel coding (SSCC) and deep joint source channel coding (DeepJSCC) are inherently limited in meeting these demands. These approaches either waste bandwidth by reconstructing human-readable data that is unnecessary for control tasks or provide uniform protection to all data regardless of its control relevance. To overcome these challenges, this study proposes a novel semantic polar coding framework termed POLAR-CODE. The proposed approach integrates semantic extraction and channel coding into a unified end-to-end architecture optimized for control-oriented communication. At its core, POLAR-CODE employs a task-driven semantic Transformer to extract compact, control-relevant representations, such as actuator signals and safety margins. It combines them with a semantics-aware Polar coding scheme. A key innovation is the Gradient-Guided Polarization (GGP) mechanism, which dynamically adjusts bit reliability based on the semantic importance of encoded features. Extensive simulations of cooperative vehicle platooning scenarios using vehicle-to-everything (V2X) channel models demonstrate the effectiveness of the approach. Results show that POLAR-CODE achieves approximately 97% payload reduction, improves lane-keeping stability by approximately 42%, and reduces emergency-braking collisions by approximately 81% compared to state-of-the-art DeepJSCC methods under stringent reliability and low signal-to-noise ratio (SNR) conditions.

1. Introduction

1.1. Background

Wireless communication plays a vital role in supporting perception sharing, cooperative decision-making, and coordinated control between vehicles and infrastructure in autonomous vehicle systems [1]. Communication networks are part of the control loop and are no longer just a supporting service as autonomous driving technologies continue to develop. Highly reliable, low-latency information exchange is needed between distributed agents for safety–critical operations such as collision avoidance, platoon stabilization, lane merging, and emergency braking [2]. Ultra-Reliable Low-Latency Communication (URLLC) was added to 5G networks for applications with low-latency requirements, including vehicle applications. All of these, however, demand communication capabilities beyond what current systems can provide: increased traffic density, multi-sensor integration, collective perception, and fully autonomous driving [3]. 6G networks are thought to be intelligent communication networks that simultaneously enable sensing, localization, and control capabilities [4]. To support cooperative driving operations in future autonomous transportation systems, sub-millisecond latency and near-perfect reliability are necessary [5]. In cooperative operations, vehicles must share their intentions, trajectories, and hazard information within very short time windows to avoid instability and the propagation of collisions [6]. Communication delays or packet losses can have a significant impact on a vehicle's performance, safety, and control [7]. To meet these challenges, 6G networks include enabling technologies like terahertz communication, visible-light communication, edge intelligence, and integrated sensing and communication (ISAC) [8,9]. Therefore, the development of ultra-reliable, low-latency communication has emerged as a critical need for next-generation autonomous transportation systems for collective perception, cooperative navigation, and intelligent traffic management.

Several complementary technologies support this vision. URLLC can deliver ultra-reliable, low-latency links for mission-critical communication systems [10]. Vehicle-to-Everything (V2X) communication enables the exchange of information among vehicles, infrastructure, pedestrians, and networks, whereas ISAC combines sensing and

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communication capabilities to improve situational awareness [11]. To enable efficient and reliable communications, edge intelligence and Reconfigurable Intelligent Surfaces are further enhanced [12]. In the area of semantic communication, the emphasis is on conveying task-relevant information rather than raw data by conveying the semantics of the message rather than the number of bits [13]. Joint source-channel coding is also a way to combine compression and error protection to enhance communication robustness in the presence of distortion and errors, while meeting tight latency requirements [14]. These technologies are the basis for future intelligent vehicular communication systems [15].

1.2. Problem Statement

Although great strides have been made in semantic communication, URLLC, edge intelligence, and 6G-enabled V2X systems, several critical issues remain to be addressed. Most semantic communication frameworks focus on semantic reconstruction and task correctness, without considering channel coding, communication reliability, and control needs. As with the former, existing URLLC solutions focus on optimising the decoder, resource allocation, and network intelligence, but typically assume that the channel coding method remains fixed, which limits performance in highly dynamic vehicular scenarios.

Communication performance can be expressed using three metrics for autonomous vehicular networks: end-to-end latency (L), packet delivery reliability (R), and bandwidth utilization (B). Current communication systems optimize for minimizing the block error rate and do not differentiate between features based on their importance to control. In scenarios with high mobility, dense traffic, and rapidly changing wireless channels, the traditional communication approach cannot simultaneously meet stringent reliability and latency requirements. Moreover, when communication impairments occur, equal protection of all transmitted information results in inefficient bandwidth use and causes the closed-loop control performance to be unstable. Thus, designing a communication system that concurrently minimizes latency, reliability, and bandwidth usage while optimizing protection of control-critical semantic information is a key challenge.

1.3. Novelty

Considering the above limitations, this work presents POLAR-CODE, a semantic-aware communication paradigm for URLLC in the autonomous vehicular network. This framework combines task-oriented semantic encoding with adaptive Polar channel coding to simultaneously optimize communication efficiency, reliability, and control performance. The key idea of the proposed framework is the Gradient-Guided Polarization (GGP) mechanism to establish a direct correlation between semantic importance and Polar-code bit-channel reliability. The method differs from traditional separated source-channel coding and Deep Joint Source-Channel Coding (DeepJSCC) methods, which assign the same error protection level to all semantic information, and dynamically assign error protection levels based on the error sensitivity of semantic information. This means that safety-related information, such as braking intentions, headway control, and cooperative driving information, is better protected than less important data. Moreover, the proposed framework explicitly links the reliability of the communication layer with the stability of the closed-loop control, enabling a unified, semantic-driven communication and control design for next-generation 6G V2X systems.

1.4. Contributions

The main contributions of this study are:

- A novel POLAR-CODE framework is proposed for ultra-reliable low-latency communication (URLLC) in sixth generation (6G) autonomous vehicle networks by integrating semantic-aware communication with adaptive Polar channel coding.
- A task-oriented semantic encoding mechanism is developed to extract control-relevant information and prioritize safety–critical semantics for cooperative driving applications.
- A Gradient-Guided Polarization (GGP)- based adaptive bit-reliability strategy is introduced to assign unequal error protection based on semantic importance, thereby improving transmission robustness.
- A unified communication and control design is established by linking semantic transmission with control-level metrics, such as lane stability and safety performance.
- Extensive experiments under realistic vehicle-to-everything (V2X) channel conditions demonstrate significant payload reduction, improved lane-keeping stability, and reduced emergency braking incidents at low signal-to-noise ratio (SNR).

1.5. Paper Organization

The rest of the paper is followed by Section 2, which discusses the recent literature review on the proposed topic. Section 3 provides a detailed description of the proposed framework. Section 4 delineates the experimental configuration, results and discussion. Section 5 presents the study's conclusion.

2. Related Works

2.1. Semantic Communication for Intelligent Vehicular Networks

In this regard, a new paradigm of communication, semantic communication, is gaining consideration for future wireless systems, focusing on the delivery of task-relevant information rather than bit recovery. The theoretical groundwork for semantic and task-oriented communications was laid by Gündüz et al. [16], who combined information-theoretic principles with deep learning-based semantic representations. Similarly, Xie et al. [17] proposed a DeepSC that was shown to be useful for robust communication in low-SNR scenarios through a Transformer-based semantic encoding. Beyond-5G networked vision was also introduced by Fernandes et al. [18], who proposed AI-enhanced semantic extraction and goal-oriented transmission mechanisms, and by Feng et al. [19], who adopted the concept of semantic communication in autonomous driving networks with the aid of edge intelligence and cooperative perception. Both approaches limit communication overhead and enhance task performance, but they tend to focus primarily on semantic reconstruction accuracy and bandwidth efficiency. The reliability requirements of safety-critical vehicular control applications have not been explicitly accounted for in most existing research, and there is no mechanism to provide semantic-aware unequal error protection. In addition, the correlation between the reliability of physical-layer channel coding and its semantic importance has not been well studied.

2.2. URLLC and 6G-Enabled Vehicular Communication

URLLC is now a key enabler for highly reliable, low-latency autonomous vehicle applications. To enable scalable and reliable vehicular communication, Noor-A-Rahim et al. [20] explored machine-learning-based 6G V2X architectures. For autonomous vehicles, Revathi [21] proposed an integrated framework combining 6G communication, artificial intelligence, blockchain, and edge computing to minimize communication latency. In the same way, Pasha et al. [22] and Osama et al. [23] explored URLLC requirements, enabling technologies, and scenarios for future 5G/6G wireless networks.

The studies show that, with the use of advanced networking technologies, there are significant improvements in communication performance. But most solutions focus on optimizing networks to support traditional communication paradigms, manage resources, or minimize

Table 1
Comparative Analysis of Related Works.

Ref.	Category	Main Focus	Key Strength	Major Limitation
[16]	Semantic Communication	Semantic and task-oriented communication	Improved task efficiency and semantic transmission	Limited reliability-aware channel coding
[17]	Semantic Communication	DeepSC semantic transmission	Robust performance under low SNR	High training complexity and computational cost
[20]	6G V2X Communication	ML-enabled intelligent vehicular communication	Improved reliability and scalability	Deployment and standardization challenges
[18]	Semantic Communication	AI-based semantic extraction and transmission	Significant bandwidth reduction	Lack of semantic reliability protection
[21]	6G Autonomous Driving	AI/ML and blockchain-assisted vehicular coordination	Ultra-low latency communication	High complexity and scalability issues
[19]	Semantic Communication for AVs	Edge-intelligence-assisted semantic communication	Reduced transmission overhead	Limited channel coding integration
[27]	Channel Coding for URLLC	Universal decoding techniques	Enhanced error correction performance	Equal protection of transmitted information
[26]	Network Intelligence	Network Digital Twin (NDT)-based optimization	Improved network planning and management	High computational overhead
[24]	Edge AI for URLLC	Edge AI-enabled communication architecture	Reduced latency and improved efficiency	Resource constraints and security concerns
[25]	Edge Computing for 6G	Edge-assisted URLLC framework	Enhanced responsiveness and reliability	Interoperability and deployment challenges
[22]	URLLC Networks	AI-assisted low-latency communication	Higher throughput and reliability	Limited practical validation
[23]	5G/6G URLLC	Enabling technologies and architectures	Addresses latency and reliability requirements	Limited scalability and real-world deployment
This	**Semantic-Aware URLLC**	**Task-oriented semantic encoding with Gradient-Guided Polarization**	**Joint optimization of semantic importance, channel reliability, and control stability**	**Addresses identified research gap**

latency at the network level. Consequently, they are not explicitly used to enhance communication reliability in safety-critical vehicular control applications by leveraging semantic importance information.

2.3. Edge Intelligence and AI-Driven Communication Architectures.

The promise of edge intelligence has been recognized as an effective way to reduce communication delays and enhance real-time decision-making. Gomathi et al. [24] presented an Edge AI-based 6G architecture based on intelligent resource allocation and adaptable network management to enable URLLC. Adhikari and Hazra [25] studied the synergy between 6G communication and edge computing for low-latency applications. Besides, Tran et al. [26] proposed a Network Digital Twin (NDT)- based network management framework for intelligent optimization and predictive control in future communication systems. Though these strategies increase the network's responsiveness and efficiency, they primarily address offloading computations, network orchestration, and resource optimization. The importance of semantic information, reliability of communication and vehicular control performance are not explicitly considered. Moreover, deployment complexity and computational overhead remain critical issues in large-scale vehicular environments.

In highly dynamic wireless environments, reliable channel coding remains a key requirement for meeting URLLC requirements. Gautam et al. [27] examined code-agnostic decoding methods for 6G URLLC systems, including Automorphism Ensemble Decoding, GRAND, Ordered Statistics Decoding, Belief Propagation Decoding, and Bit-Flipping Decoding. These techniques enhance the channel's decoding performance and error-correction capability under difficult channel conditions. While these techniques are successful, current channel coding methods generally consider only channel characteristics when determining the amount of protection to allocate and do not account for the semantics of the transmitted information. Thus, all control-critical and non-critical data are protected at the same level, resulting in inefficient use of communication resources in autonomous vehicular applications.

2.4. Compare and contrast findings with other studies and identify gaps

The most important characteristics and the limitations of the studies are summarized in Table 1. The existing semantic communication schemes efficiently compress information and convey the task, but fail to provide reliable semantic protection, unlike VLC, URLLC, and 6G vehicular communication solutions, which are mainly based on reducing latency and optimizing the network without considering semantic importance in channel coding design. Edge intelligence architectures are designed to enhance computational efficiency without necessarily directly linking communication reliability to control stability. For now, existing channel coding techniques improve transmission reliability but do not account for the varying importance of semantic information.

Thus, there is a critical research gap: a unified framework that seamlessly combines semantic communication, adaptive channel coding, and control-aware reliability optimization. This is described in Table 1. Specifically, current work does not provide a mechanism for dynamically providing unequal error protection based on semantic importance and does not explicitly correlate communication-layer reliability with closed-loop autonomous vehicle control performance. To overcome these challenges, this work proposes the POLAR-CODE framework, which combines task-oriented semantic encoding with Gradient-Guided Polarization-based adaptive Polar coding to enable reliable, low-latency, control-aware communication in 6G autonomous vehicular networks.

To address sensing occlusions in connected autonomous vehicles, Xiao et al. [28] proposed an information-sharing framework based on perception. Wang et al. [29] proposed an Interactive Siamese Network for a roadside multi-vehicle tracking system, which improves tracking consistency in complex traffic scenarios. Wang et al. [30] proposed a Synchronous Spatiotemporal Graph Transformer to model the spatial–temporal relationships in data for accurate traffic prediction. Wang et al. [31] proposed a knowledge-guided self-learning control strategy that enhances the stability of mixed vehicle platoons under communication delays. He et al. [32] presented a distributed model predictive control (MPC) scheme for heterogeneous connected autonomous vehicle (CAV) platoons with Markovian switching communication topologies.

These studies improve perception sharing, vehicle tracking, traffic prediction, and cooperative control, but they fail to combine semantic-aware communication with adaptive channel coding or to use unequal error protection based on semantic importance. Also, the stability of closed-loop control with respect to communication reliability has not been fully studied. Thus, the proposed POLAR-CODE framework overcomes these issues by leveraging semantic encoding with a Transformer

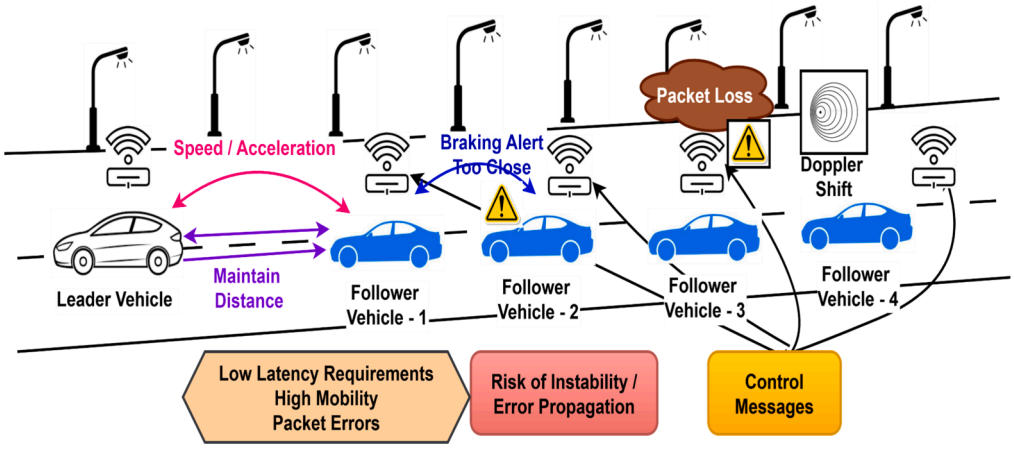


Fig. 1. Cooperative Vehicle Platooning Over Semantic-Aware V2X Communication.

and Gradient-Guided Polarization-enabled Polar coding for reliable, control-aware 6G vehicular communications.

Emerging developments in 6G vehicular networking have further underscored the need for reliable communication, cooperative perception, and intelligent task management in autonomous driving and transportation systems. Recent 6G vehicular networking progress has also clearly demonstrated the need for reliable communication, cooperative perception, and intelligent task management in autonomous driving and transportation systems. Emerging developments in 6G vehicular networking have further emphasized the need for reliable communication, cooperative perception, and intelligent task management in autonomous driving and intelligent transportation systems. Jiang et al. [33] proposed a collaborative perception and computation offloading framework for 6G air–ground integrated networks, demonstrating that joint perception sharing and edge-assisted computing can significantly improve situational awareness, computational efficiency, and communication performance in highly dynamic vehicular environments. Peng et al. [34] introduced the 2FDP-BRL distributed task offloading framework for the Internet of Autonomous Vehicles (IoAV) operating under extreme weather conditions, where reinforcement learning and adaptive resource management effectively reduced service latency while improving computational resource utilization and task execution reliability. Furthermore, Jin et al. [35] developed a deep reinforcement learning-based redundant transmission scheduling strategy over Gilbert–Elliot channels, providing stability-guaranteed remote state estimation by enhancing transmission reliability and robustness under time-varying wireless channel conditions. Collectively, these studies demonstrate that integrating collaborative perception, intelligent edge computing, and reliable communication mechanisms is fundamental for next-generation 6G-enabled autonomous vehicular networks, thereby motivating the proposed semantic-aware POLAR-CODE framework for ultra-reliable, low-latency vehicular communications.

The proposed POLAR-CODE framework is distinct from existing

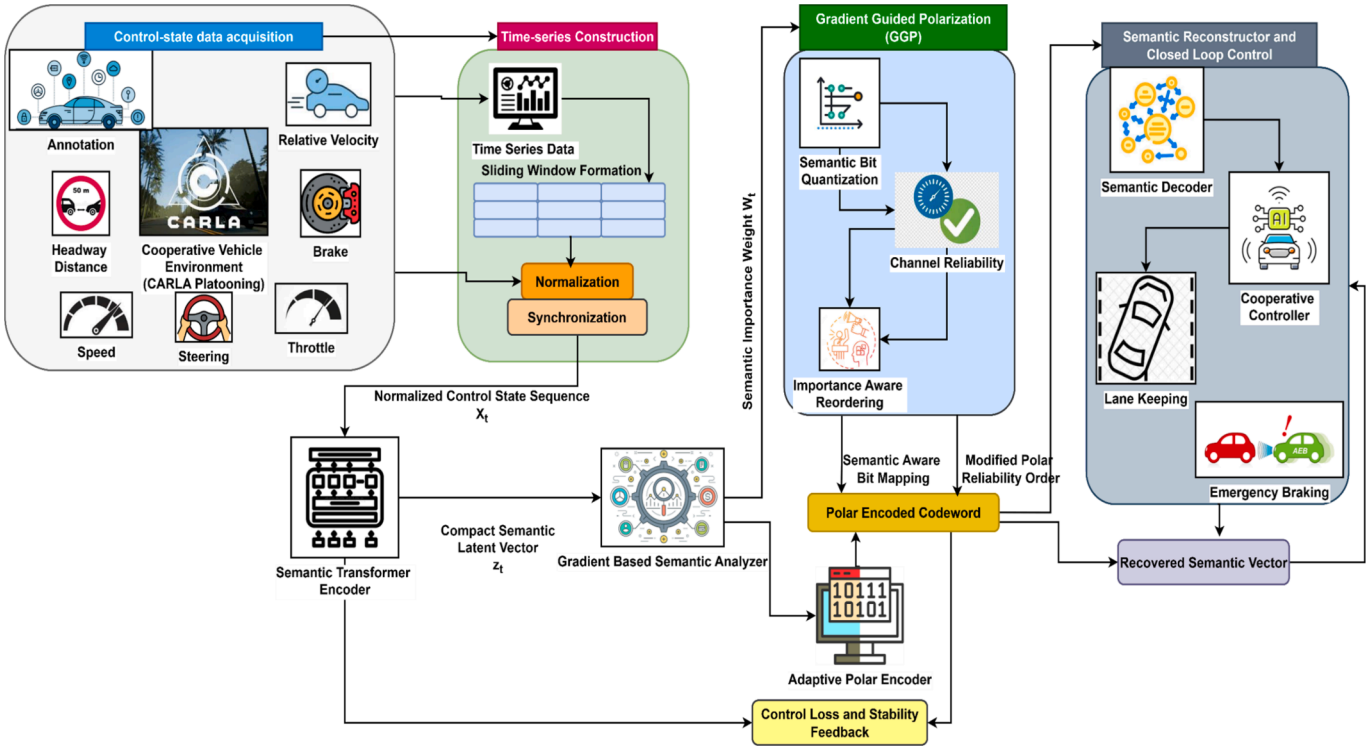


Fig. 2. Overall Architecture: POLAR-CODE End-to-End Semantic Polar URLLC Framework.

semantic communication and adaptive coding approaches through its novel Gradient-Guided Polarization (GGP) mechanism, which jointly exploits semantic importance and Polar channel reliability. Unlike DeepSC [17] and DeepJSCC-based semantic communication frameworks [36,37], which provide uniform protection to latent semantic features, the proposed method dynamically allocates unequal error protection according to control relevance. Experimental results demonstrate that POLAR-CODE achieves a payload reduction of 97–98.1%, significantly outperforming DeepJSCC (40–48%), while maintaining substantially lower BLER under low-SNR V2X conditions. In addition, the framework improves lane-keeping stability by approximately 42% and reduces emergency-braking collision events by approximately 81% compared with existing semantic communication baselines. Furthermore, POLAR-CODE operates with a compact 512-bit Polar code and sub-millisecond latency, thereby establishing a direct relationship between semantic transmission reliability and closed-loop autonomous vehicle control performance, a capability not addressed by existing semantic-aware communication or adaptive Polar coding schemes [17,27,36,37].

3. Proposed Methodology

Fig. 1 depicts a cooperative vehicle platooning scenario utilizing V2X communication. A lead vehicle transmits essential semantic information, including speed, acceleration, and brake intent, to several follower vehicles to ensure safe inter-vehicle distance. Wireless channel impairments, such as packet loss and Doppler effects from high mobility, might hinder timely information sharing, potentially leading to spacing errors and increased collision risk. This number underscores the necessity for semantic-aware, reliability-driven communication systems to ensure reliable and secure autonomous platooning. A semantic encoder based on Transformers is then used to mine a small, priori latent representation that captures control-relevant semantics pertinent to cooperative driving tasks and represses non-relevant information. Gradient-based sensitivity analysis is used to determine the contribution of each latent component to controlling stability and safety goals, while accounting for the non-uniform significance of semantic features. Based on this important information, a Gradient-Guided Polarization mechanism dynamically reallocates the reliability ranking of Polar code bit-channels, allowing unequal error protection based on semantic relevance.

The proposed POLAR-CODE framework adheres to the end-to-end, task-driven workflow needed to address ultra-reliable, low-latency control in 6G autonomous vehicle networks. As shown in Fig. 2, the workflow starts by obtaining vehicle control and state information from a CARLA-based autonomous driving simulation, with all variables such as speed, steering angle, throttle, brake, and vehicle pose continuously collected. These signals are converted to time-series representations and preprocessed to capture the temporal driving dynamics. The semantically conscious bitstream is then encoded using an adaptive Polar encoder and relayed over a realistic V2X wireless channel under severe URLLC constraints. The control-relevant information is reconstructed at the receiver via Polar decoding and semantic reconstruction and is directly used to perform cooperative control functions such as platoon stabilization, lane maintenance, and emergency braking. Lastly, the framework is tested using communication-level and control-level measures to demonstrate its efficiency compared to traditional coding schemes.

The entire POLAR-CODE framework is shown in Fig. 2, starting with the acquisition of CARLA-based vehicle control-state data and time-series preprocessing. These processed control signals are then mapped to compact semantic representations with an encoder based on the Transformer model. A Gradient-Guided Polarization (GGP) module estimates semantic importance and assigns reliable Polar sub-channels adaptively with respect to semantic importance. The semantic-aware URLLC communication architecture reconstructs the semantic data transmitted via the V2X channel, enabling cooperative control functions such as lane keeping, platoon stabilization, and emergency braking.

3.1. Control-State Data Acquisition

In the proposed framework for POLAR-CODE, the communication process begins by obtaining vehicle control and state data from the CARLA Autopilot Images dataset [38], which is generated using a simulated autonomous driving platform. The dataset provides high-frequency measurements of the ego-vehicle's states, simulating realistic driving conditions. At each discrete time step, the system can transition t, the ego vehicle state is represented by a control-state vector: $x_t = \{v_t, a_t, \delta_t, \tau_t, b_t, p_t, \psi_t\}$; where: v_t denotes vehicle speed, a_t is longitudinal acceleration, δ_t represents steering angle, τ_t and b_t corresponding to throttle and brake control inputs, respectively, $p_t = (x_t, y_t)$ is the vehicle position, and ψ_t indicates the yaw (heading) angle.

To incorporate the temporal dependencies that are crucial for performing the control and stability analyses, a window of a fixed length consisting of consecutive control-state vectors is defined: $X_t = \{x_{t-L+1}, x_{t-L+2}, \cdots..x_t\}$; here: $L-$ the window length. This time-series representation will serve as input for the semantic encoder in subsequent stages. The obtained data directly correlates with safety-critical driving behaviors such as lane keeping, speed management, and braking reactions. The emphasis on such variables, rather than sensor measurements, will allow the suggested system to efficiently communicate information directly associated with cooperative control goals, especially for ultra-reliable low-latency communication.

The acquisition and organization of control-state information, the backbone of the suggested POLAR-CODE framework, are described in Algorithm 1. The algorithm operates in one of the CARLA-based platooning setups, periodically gathering ego-vehicle movement and control variables, including speed, acceleration, steering angle, throttle, brake, position, and heading. These variables represent momentary driving behavior, which directly impacts vehicle stability and safety. The algorithm also calculates inter-vehicle measures by consulting the state of the previous vehicle, using which the headway distance and relative velocity are determined to support cooperative platooning. These amounts are fundamental for ensuring safe clearance and for braking in time to avoid platoon instability.

Algorithm 1: Control-State Data Acquisition

Input:
CARLA simulator environment, sampling interval Δt, window length L
Output:
Time-series control-state sequence $\mathbf{X}_t$
Algorithm:
Initialize ego vehicle and platoon configuration in CARLA
For each time step t:
a. Acquire ego-vehicle states $v_t, a_t, \delta_t, \tau_t, b_t, \mathbf{p}_t, \psi_t$
b. Obtain the leader vehicle position $\mathbf{p}_t^{\text{lead}}$ and speed v_t^{lead}
c. Compute headway distance
$d_t = \|\mathbf{p}_t - \mathbf{p}_t^{\text{lead}}\|$
d. Compute relative velocity
$\Delta v_t = v_t - v_t^{\text{lead}}$
e. Form platooning-aware control vector
$\tilde{\mathbf{x}}_t = [\mathbf{x}_t, d_t, \Delta v_t]$
Aggregate the last L control vectors to form
$\mathbf{X}_t = \left[\tilde{\mathbf{x}}_{t-L+1}, \cdots, \tilde{\mathbf{x}}_t\right]$
Output $\mathbf{X}_t$ for semantic encoding

A single control vector that represents the local dynamics and interaction context is then formed by combining the ego-state and the cooperative variables. Lastly, recurring control vectors are concatenated to form fixed-length time-series windows, thereby maintaining temporal driving patterns. This is a structured, task-based representation that ensures only information critical to control is passed to subsequent stages of semantic encoding and communication within URLLC constraints.

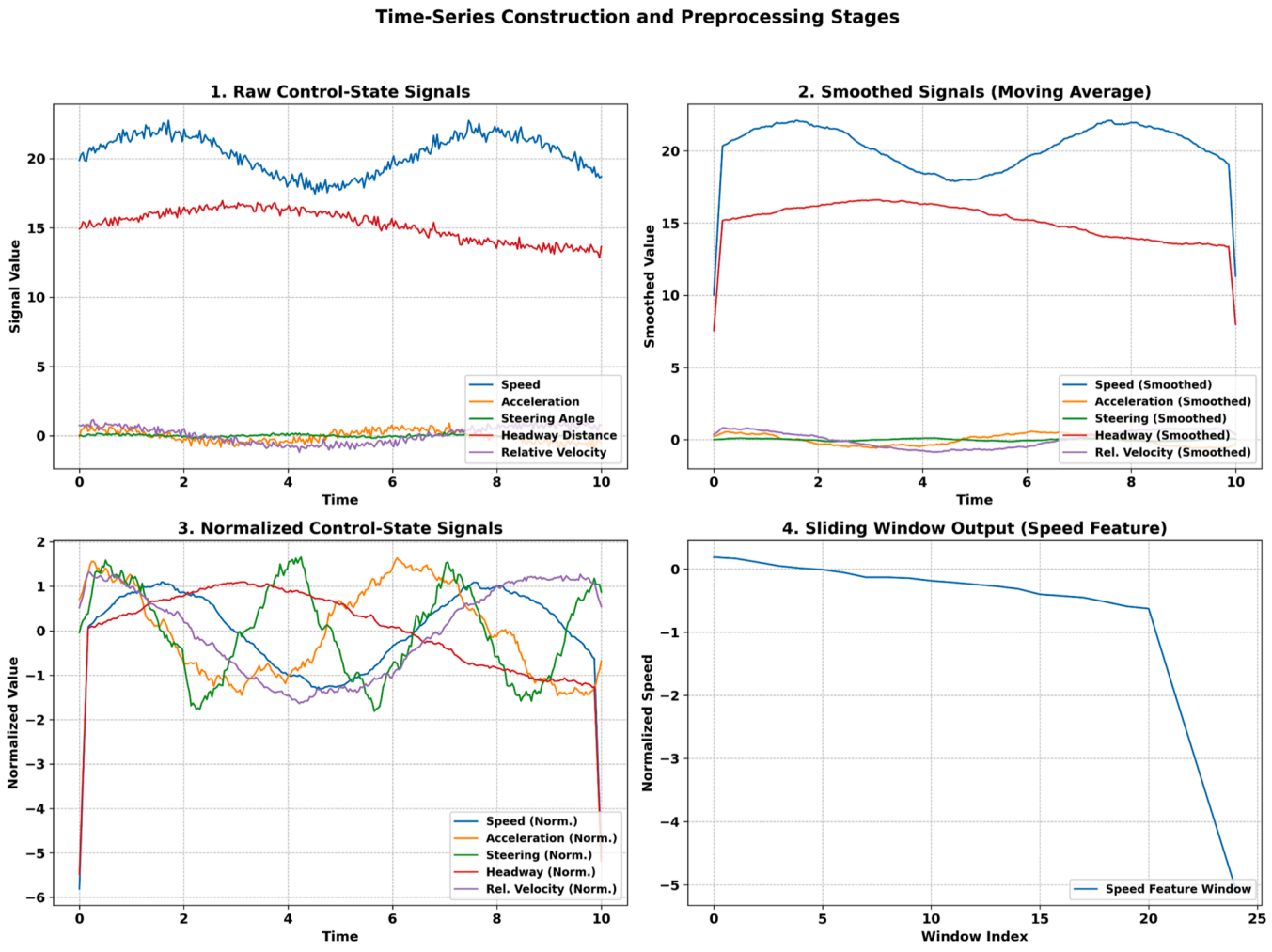


Fig. 3. Time-Series Preprocessing for Semantic Encoding.

3.2. *Time-Series Construction and Preprocessing*

Following the learning of platooning-conscious control-state vectors, the suggested framework builds temporally organized inputs to represent the dynamic development of vehicle motion and the interaction between vehicles. Let the control-state vector t be denoted as $\widetilde{x}_t = \{v_t, a_t, \delta_t, \tau_t, b_t, p_t, \psi_t, d_t, \Delta v_t\}^T \in \mathbb{R}^D$, where d_t is the headway distance, and Δv_t is the relative speed in relation to the platoon leader. These variables collectively define the ego-vehicle dynamics and the inter-vehicle safety limitations. Their presence guarantees that the following semantic representation is closely related to control stability and cooperative safety, which is essential to ultra-reliable low-latency communication. To capture temporal dependencies in vehicle control, a fixed-size sliding window of size L is constructed as in Eq. (1),

$$X_t = \begin{bmatrix} \widetilde{X}_{t-L+1} \\ \widetilde{X}_{t-L+2} \\ .. \\ .. \\ .. \\ \widetilde{x}_t \end{bmatrix} = \left[\widetilde{x}_{t-k}\right]_{k=0}^{L-1} \in \mathbb{R}^{L\times D} \tag{1}$$

The representation given in Eq. (1) preserves the chronological development of the control actions and the relative movement successively. The proposed system will allow the semantic encoder to acquire motion trends, braking intent, and spacing regulation behavior, as it operates on short temporal histories rather than instantaneous snapshots. This time, consciousness specifically enhances resilience against packet loss because the decisions made in control are based on patterns rather than single values. To reduce noise caused by sensing uncertainty and simulation artifacts, all the control variables are smoothed with a temporal moving-average filter represented in Eq. (2)

$$\left.\begin{aligned} \widehat{x}_t^{(i)} &= \frac{1}{K}\sum_{k=0}^{K-1} x_{t-k}^{(i)} \\ &= \frac{x_t^{(i)} + x_{t-1}^{(i)} + \cdots.. + x_{t-K+1}^{(i)}}{K}, i = 1, 2\cdots..D \end{aligned}\right\} \tag{2}$$

Here, $x_t^{(i)}$ denotes the $i-th$ of the raw control vector, and K is the smoothing window length. High-frequency variations, which are insignificant for control semantics, are suppressed by this kind of filtering, whereas the dominant trends, such as deceleration patterns and spacing shifts, are preserved. Consequently, more stable semantic encoder inputs will enhance the accuracy of gradient-based importance estimation, which will be employed in the Gradient-Guided Polarization mechanism. After noise reduction, feature normalization is used to ensure uniformity across control variables of different scales. All the filtered variables are normalized using the expression defined in Eq. (3),

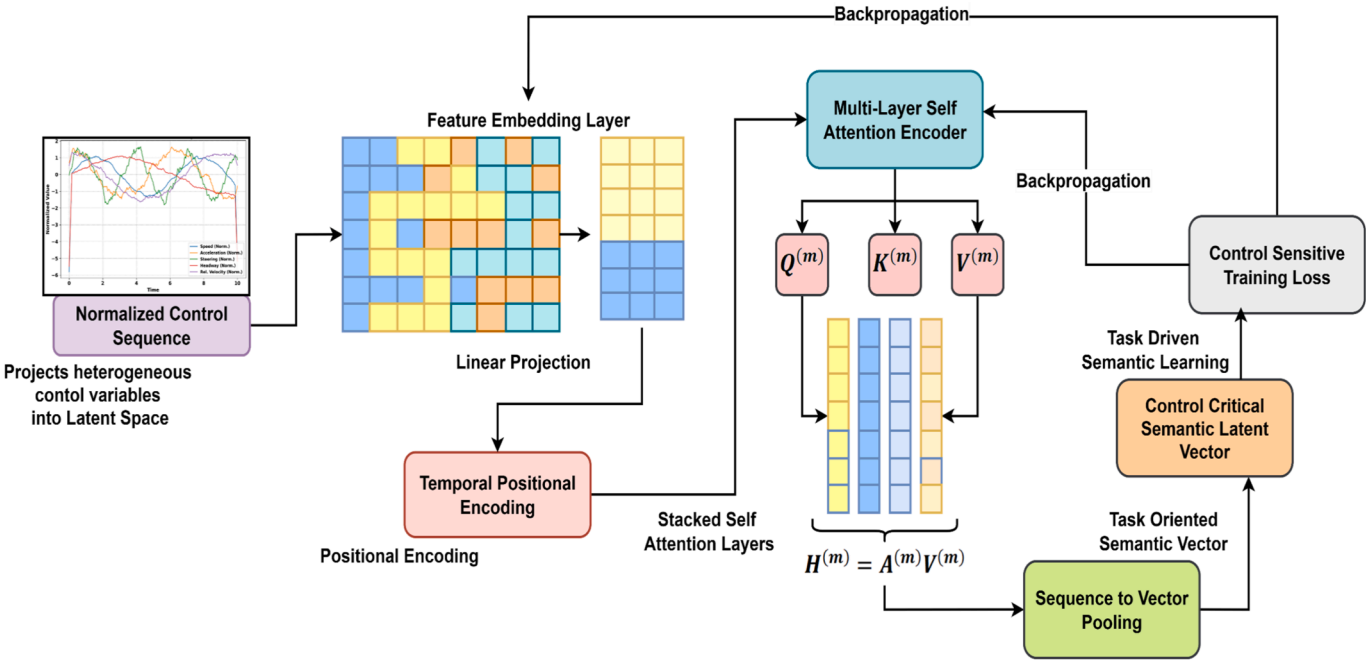


Fig. 4. Task-oriented semantic encoding module of the proposed POLAR-CODE framework.

$$\left.\begin{aligned} \bar{x}_t^{(i)} &= \frac{\hat{x}_t^{(i)} - \mu_i}{\sigma_i} \\ \mu_i &= E\left[\hat{x}^{(i)}\right] \\ \sigma_i &= \sqrt{\hat{x}^{(i)}} \end{aligned}\right\} \tag{3}$$

Here, μ_i and σ_i represent the standard deviation and average of the $i-th$ variable calculated on the training set. Normalization will help ensure that no variable, e.g., speed or distance, dominates the learning process due to scale differences. This is especially significant in semantic communication, where equal feature magnitudes allow the semantic encoder to weight by task relevance rather than by numeric scale, thereby enhancing the unequal protection effectiveness of errors. Lastly, all normalized control variables are synchronized in time at a given sampling period Δ_t in which the resultant input sequence is determined as $X_t = \{\bar{x}_{t-L+1}, \bar{x}_{t-L+2}, \cdots \bar{x}_t\}$, $t = n\Delta t$, $n \in \mathbb{Z}^+$. This synchronization ensures that ego-vehicles and platoons are aligned, which is key to cooperative control activities.

Fig. 3 illustrates the progressive conversion of unrefined platooning control signals into a temporally organized and noise-resistant representation. Temporal smoothing efficiently mitigates high-frequency variations while maintaining primary motion patterns, including braking and spacing regulation. Feature normalization standardizes data across different scales, averting the predominance of high-magnitude values. The resultant sliding-window tensor captures short-term motion dynamics and inter-vehicle interactions, enabling the semantic encoder to identify control-critical temporal patterns and providing dependable, latency-sensitive transmission in URLLC-enabled cooperative driving contexts. The time-series tensor resulting from this serves as input to the semantic Transformer encoder, providing a consistent, noise-robust, and temporally coherent representation. With such a data structure, the proposed POLAR-CODE framework will allow conveying only control-critical temporal semantics and guaranteeing them high URLLC-constrained protection.

3.3. Task-Oriented Semantic Encoding

After time-series construction and preprocessing, the normalized sequence of the control state $X_t \in \mathbb{R}^{L\times D}$ is sent to a task-oriented semantic encoder implemented as a Transformer architecture. This encoder is not aimed at reconstructing the original control signals, but at deriving a small latent representation that captures only the semantics most likely to be useful for cooperative vehicle control and safety-critical decision-making. As shown in Fig. 4, the normalized control-state sequence is mapped into a latent embedding space, enhanced with positional encoding, and analyzed by stacked self-attention layers to emphasize safety-critical temporal patterns. The concluding sequence is temporally aggregated into a concise semantic vector that encapsulates high-level control intent, which is refined using a control-sensitive loss and subsequently used to guide gradient-guided Polar coding. The projection of each time step of the sequence is first projected into a higher-dimensional embedding space defined as in Eq. (4),

$$\left.\begin{aligned} z_l &= W_e \bar{x}_{t-L+\ell} + b_e; l = 1, 2, \cdots L \\ Z &= \begin{bmatrix} z_1 \\ z_2 \\ .. \\ .. \\ z_L \end{bmatrix} \in \mathbb{R}^{L\times d_e} \end{aligned}\right\} \tag{4}$$

where $\bar{x}_l$ constitutes the normalized control vector, $W_e \in \mathbb{R}^{d_e\times D}$ and b_e are parameters that can be learned, and d_e is the embedding dimension. This transformation converts heterogeneous control variables (ex, speed, headway distance, and relative velocity) into a single latent space that can undergo attention-based processing.

The autonomous vehicle's information is multivariate temporal sequences with complex dependencies among the variables of speed, acceleration, steering angle, headway distance, and relative velocity, so the Transformer architecture was adopted to extract control-semantic information. The self-attention mechanism enables the model to learn short- and long-range temporal dependencies, which are crucial for understanding safety-critical driving behaviors such as braking intent, lane correction, and platoon stabilization. This helps create compact semantic representations that contain task-relevant control information and avoid redundant data. The selection of the gradient magnitude for

estimating semantic importance was based on the direct relationship between the gradient magnitude and the sensitivity of the control objective to each latent semantic feature. The greater the gradient's magnitude, the more it affects tracking accuracy, stability maintenance, and collision avoidance. In this way, gradient-based importance estimation provides a principled means of measuring semantic relevance. It facilitates the adaptive allocation of communication resources based on each semantic feature's contribution to overall control performance.

The selection of polar coding was made because it is well-suited for short-block-length Ultra-Reliable Low-Latency Communication (URLLC) scenarios and provides a reliable ordering of synthesized bit-channels. This property complements the proposed Gradient-Guided Polarization mechanism, in which the control-critical semantic bits are transmitted through the most reliable channel. At the same time, less critical information is given lesser protection. Polar codes are well suited for semantic-aware communication in autonomous vehicle networks, as they offer a good trade-off between reliability, decoding complexity, latency, and unequal error protection capability compared with other short-block-length coding schemes. By combining these variables, the encoder acquires an association among longitudinal, lateral, and cooperative dynamics, which is critical for control-based semantic abstraction. Positional encoding is introduced into the encoded sequence to capture temporal order and motion evolution, as given in Eq. (5).

$$\left.\begin{aligned} H^{(0)} &= Z + P \\ P_{\ell,j} &= \sin\left(\frac{\ell}{10000^{\frac{2j}{d_e}}}\right) or \cos\left(\frac{\ell}{10000^{\frac{2j}{d_e}}}\right) \end{aligned}\right\} \tag{5}$$

where P inserts time step data into the sequence model. This operation will allow the model to discriminate between early and late control actions within the window, enabling it to detect trends such as accelerating deceleration or decreasing headway. Maintaining the time sequence is important to the process of identifying the control purpose, particularly in emergency brake and lane stabilization operations. Stacked self-attention layers are used to perform core semantic extraction. For each layer m, the latent representation is updated with the help of the mathematical expression using Eq. (6),

$$\left.\begin{aligned} Q^{(m)} &= H^{(m-1)} W_Q^{(m)} \\ K^{(m)} &= H^{(m-1)} W_K^{(m)} \\ V^{(m)} &= H^{(m-1)} W_V^{(m)} \\ A^{(m)} &= softmax\left(\frac{Q^{(m)} K^{(m)T}}{\sqrt{d_k}}\right) \\ H^{(m)} &= A^{(m)} V^{(m)} \end{aligned}\right\} \tag{6}$$

where $Q^{(m)}$, $K^{(m)}$, $V^{(m)}$ are query, key, and value matrices, d_k is the key dimension, and $A^{(m)}$ represents the weights of attention. With this mechanism, the encoder will prioritize time steps and variables that have the greatest impact on the controller's ability to maintain stability, including a sudden decrease in the distance between headways or a sudden change in relative velocity. Such preferential aggregation silences redundant steady-state information and accentuates safety-critical transitions, which is directly useful to semantic compression. The sequence-level semantic representation is subsequently summarized into a small latent vector after several attention layers through temporal pooling defined in Eq. (7),

$$s_t = \frac{1}{L}\sum_{\ell=1}^{L} H_\ell^{(M)} = POOL\left(H^{(M)}\right) \in \mathbb{R}^{d_s} \tag{7}$$

where $H^{(M)}$ is the output of the last Transformer layer and $d_s \ll D \times L$.

This latent semantic vector represents high-level control intent, e.g., platoon spacing, lane deviation correction, or emergency braking. Its simplicity is important for reducing communication payload and preserving task-relevant meaning. The encoder is task-trained by minimizing a control-sensitive loss function: $\mathscr{L} = \lambda_1 . \mathscr{L}_{track} + \lambda_2 . \mathscr{L}_{stability} + \lambda_3 . \mathscr{L}_{safety}$; in which tracking loss discourages errors in speed and distance, stability loss represents oscillations or string instability, and safety loss focuses on collision avoidance and braking response. This goal ensures that the learned semantics are consistent with control performance rather than signal reconstruction accuracy. The latent representation s_t is therefore the one obtained naturally, which places more emphasis on the variables and time steps that affect safety and stability, and is thus well suited to gradient-guided Polar coding. In general, this task-based semantic encoding step converts the raw control sequences into a concise, control-critical semantic representation that directly facilitates importance-aware channel protection and ultra-reliable low-latency cooperative driving.

Algorithm 2 summarizes the POLAR-CODE framework's task-oriented semantic encoding process. The temporal window loop embeds each normalized control-state vector into a shared latent space, enabling the modeling of ego-vehicle dynamics and platooning interactions together. To recognize changing control patterns, such as lane corrections and emergency braking, positional encoding maintains temporal order. The stacked self-attention loops enhance the representation by highlighting control stability and safety-relevant time steps and variables and reducing redundant steady-state information. Temporal pooling compresses the refined sequence into a semantic vector that represents high-level control intent rather than signal values. The final while loop optimizes control-aware objectives to promote task-oriented learning and ensure that the extracted semantics support reliable, low-latency cooperative driving within the semantic-polar coding framework.

Algorithm 2: **Task-Oriented Semantic Encoding**

Input:
Normalized control-state sequence $X_t = \left[\bar{x}_{\{t-L+1\}}, \cdots, \bar{x}_t\right]$
Embedding dimension d_e, number of Transformer layers M
Output:
Compact semantic latent vector s_t
Initialize empty embedding matrix Z
for $\ell = 1 to L$ do
$z_\ell \leftarrow W_e \bar{x}_{\{t-L+\ell\}} + b_e$
append $z_\ell to Z$
end for
$H^{(0)} \leftarrow Z + P$// positional encoding
for $m = 1 to M$ do
$Q \leftarrow H^{(m-1)} W_Q^{(m)}$
$K \leftarrow H^{(m-1)} W_K^{(m)}$
$V \leftarrow H^{(m-1)} W_V^{(m)}$
$A \leftarrow softmax\left(\frac{(QK^T)}{\sqrt{d_k}}\right)$
$H^{(m)} \leftarrow AV$
end for
$s_t \leftarrow 0$
for $\ell = 1 to L$ do
$s_t \leftarrow s_t + H_\ell^{(M)}$
end for
$s_t \leftarrow \left(\frac{1}{L}\right) s_t$// temporal pooling
while training loss not converged do
$\mathscr{L} \leftarrow \lambda_1 \mathscr{L}_{track} + \lambda_2 \mathscr{L}_{stability} + \lambda_3 \mathscr{L}_{safety}$
update encoder parameters via backpropagation
end while
return s_t

Semantic Importance Estimation

The control-relevant information at that time that has been seman-

tically encoded t is a latent vector that is compact $s_t = \begin{bmatrix} s_t^{(1)} \\ s_t^{(2)} \\ .. \\ .. \\ s_t^{(d_s)} \end{bmatrix} \in \mathbb{R}^{(d_s)}$ and the components correspond to abstract semantic features that represent high-level driving intent, e.g. speed control, spacing control, or emergency response. These semantic features are small; however, they do not play an equally important part in the control task. The suggested framework estimates gradient-based semantic importance relative to a task-oriented control objective to explicitly quantify their relative importance. Let the total loss of control be referred to as $\mathscr{L} \leftarrow \lambda_1 \mathscr{L}_{track} + \lambda_2 \mathscr{L}_{stability} + \lambda_3 \mathscr{L}_{safety}$ that combines punishment for tracking errors, platoon instability, and safety breaches. The sensitivity of this objective to every semantic feature is then computed by computing the gradient of the loss with respect to the latent representation (Eq. (8)):

$$g_t = \frac{\partial L}{\partial s_t} = \begin{bmatrix} \frac{\partial L}{\partial s_t^{(1)}} \\ \frac{\partial L}{\partial s_t^{(2)}} \\ \vdots \\ \frac{\partial L}{\partial s_t^{(d_s)}} \end{bmatrix} = \begin{bmatrix} g_t^{(1)} \\ g_t^{(2)} \\ . \\ . \\ . \\ g_t^{(d_s)} \end{bmatrix} \tag{8}$$

The gradient vector g_t represents the sensitivity of the control-aware loss function L with respect to the semantic latent representation s_t at time step t. Specifically, $g_t = \frac{\partial L}{\partial s_t}$ quantifies how variations in each semantic feature affect the overall control objective. Here, L denotes the total control-aware loss function, which combines tracking accuracy, platoon stability, and safety-related penalties, while $s_t \in \mathbb{R}^{d_s}$ represents the semantic latent vector extracted by the Transformer-based semantic encoder at time instant t. The parameter d_s denotes the dimensionality of the semantic latent space. The vector g_t is composed of individual gradient components $\frac{\partial L}{\partial s_t^{(1)}}$, where $s_t^{(i)}$ is the ith semantic feature is in the latent representation. Each element $g_t^{(i)} = \frac{\partial L}{\partial s_t^{(i)}}$ measures the contribution of the corresponding semantic feature to the control loss. A larger magnitude of $g_t^{(i)}$ indicates that the associated semantic feature has a stronger influence on vehicle control performance and is therefore more critical for maintaining cooperative driving stability and safety. Consequently, the gradient vector serves as the foundation for semantic importance estimation, enabling the proposed Gradient-Guided Polarization mechanism to assign higher transmission reliability to control-critical semantic information.

Each gradient component $g_t^{(i)}$ tests the extent to which a minor modification of the respective semantic attribute influences control performance. The bigger they are, the more sensitive and, therefore, the more semantically important. To have a non-negative scalar measure of importance to indicate the strength of influence independent of sign, the gradient vector is changed by using an element-wise magnitude operation, denoted in Eq. (9):

$$\alpha_i^{(t)} = \left| g_t^{(i)} \right| = \sqrt{\left(\frac{\partial \mathscr{L}}{\partial s_t^{(i)}} \right)^2}, i = 1, 2, \cdots d_s \tag{9}$$

resulting in the crude semantic importance vector $\alpha_t = \begin{bmatrix} \alpha_t^{(1)} \\ \alpha_t^{(2)} \\ .. \\ .. \\ \alpha_t^{(d_s)} \end{bmatrix}$. This formulation ensures that semantic features related to abrupt braking, rapid decreases in headway, or aggressive steering corrections are given high scores. In contrast, steady-state cruising features are given lower values.

The semantic importance score $\alpha_i^{(t)}$ represents the significance of the ith semantic feature at time step t. It is computed as the absolute magnitude of the corresponding gradient component, $\alpha_i^{(t)} = \left| g_t^{(i)} \right| = \sqrt{\left(\frac{\partial L}{\partial s_t^{(i)}} \right)^2}$. Here, L denotes the overall control-aware loss function, $s_t^{(i)}$ represents the ith element of the semantic latent vector s_t and $\frac{\partial L}{\partial s_t^{(i)}}$ measures the sensitivity of the loss function to changes in that semantic feature. The term $g_t^{(i)}$ denotes the gradient associated with the ith semantic dimension, while d_s represents the total number of semantic features in the latent space. The index $i = 1, 2, \cdots, d_s$ identifies each semantic feature individually. A larger value of $\alpha_i^{(t)}$ indicates that the corresponding semantic feature has a greater influence on control performance and should therefore receive higher transmission protection in the proposed Gradient-Guided Polarization framework. The raw importance scores are normalised to generate a semantic importance distribution with no scale-dependent bias: because the absolute magnitude of gradients can change over time and driving conditions, the raw importance scores are normalised as in Eq. (10)

$$\beta_t = \frac{\alpha_t}{\sum_{i=1}^{d_s} \alpha_t^{(i)}} = \begin{bmatrix} \frac{\alpha_t^{(1)}}{\sum_{j=1}^{d_s} \alpha_t^{(j)}} \\ \frac{\alpha_t^{(2)}}{\sum_{j=1}^{d_s} \alpha_t^{(j)}} \\ .. \\ .. \\ .. \\ \frac{\alpha_t^{d_s}}{\sum_{j=1}^{d_s} \alpha_t^{(j)}} \end{bmatrix}, \sum_{i=1}^{d_s} \beta_t^{(i)} = 1 \tag{10}$$

The normalized semantic importance vector β_t represents the relative importance of all semantic features at time step t. It is obtained by dividing each raw semantic importance score $\alpha_t^{(i)}$ by the sum of all importance scores, ensuring that the importance values are scaled to a common range. Here, $\alpha_t = \left[\alpha_t^{(1)}, \alpha_t^{(2)}, \cdots, \alpha_t^{(d_s)} \right]$ denotes the raw semantic importance vector, where $\alpha_t^{(i)}$ is the importance score of the ith semantic feature and d_s is the total number of semantic dimensions. The resulting vector $\beta_t = \left[\beta_t^{(1)}, \beta_t^{(2)}, \cdots, \beta_t^{(d_s)} \right]$ expresses the relative contribution of each semantic feature to the overall control objective. Since the normalization enforces $\sum_{i=1}^{d_x} \beta_t^{(i)} = 1$, the values of β_t can be interpreted as a probability-like distribution of semantic importance. Features with larger $\beta_t^{(i)}$ values are considered more critical for vehicle control and are therefore assigned higher protection during semantic-aware Polar coding.

The normalized version of this is the direct expression of the relative

semantic significance of each latent feature and is directly applicable to channel coding resource mapping. To enhance resilience to temporary gradient variations due to noise or abrupt maneuvering, the importance vector is smoothed in time with exponential averaging (Eq. (11)):

$$\overline{\beta}_t = (1-\eta)\overline{\beta}_{t-1} + \eta\beta_t = \begin{bmatrix} \overline{\beta}_t^{(1)} \\ \overline{\beta}_t^{(2)} \\ \cdot\cdot \\ \cdot\cdot \\ \cdot\cdot \\ \overline{\beta}_t^{(d_s)} \end{bmatrix}, 0 < \eta \leq 1. \tag{11}$$

This operation focuses more on consistent control-critical semantics without sudden redistribution of significance as caused by temporal disruptions. The resulting semantic importance-smoothed-out vector $\overline{\beta}_t$ forms a principled connection between the sensitivity of control and the reliability of communication. This step quantifies the impact of each semantic feature on control objectives by explicitly quantifying how much safer parts of such information are to be assigned equal error protection, ensuring that safety-critical information will not be lost in severe URLLC conditions.

3.5. Gradient-Guided Polarization (GGP)

The estimation of the control-aware relevance of each latent semantic dimension at time t, followed by the estimation of semantic importance done by the smoothed importance of the vector of expanded joints, written as in Eq. (12),

$$\overline{\beta}_t = \begin{bmatrix} \overline{\beta}_t^{(1)} \\ \overline{\beta}_t^{(2)} \\ \cdot\cdot \\ \cdot\cdot \\ \cdot\cdot \\ \overline{\beta}_t^{(d_s)} \end{bmatrix} = (1-\eta) \begin{bmatrix} \overline{\beta}_{t-1}^{(1)} \\ \overline{\beta}_{t-1}^{(2)} \\ \cdot\cdot \\ \cdot\cdot \\ \cdot\cdot \\ \overline{\beta}_{t-1}^{(d_s)} \end{bmatrix} + \eta \begin{bmatrix} \overline{\beta}_t^{(1)} \\ \overline{\beta}_t^{(2)} \\ \cdot\cdot \\ \cdot\cdot \\ \cdot\cdot \\ \overline{\beta}_t^{(d_s)} \end{bmatrix}, 0 < \eta \leq 1. \tag{12}$$

This vector is a temporally constant approximation of the strength of each semantic aspect relative to the control goal and serves as the control cue for channel polarization. The semantic latent vector s_t is then converted to a quantized binary stream that can be channel-encoded. This procedure can be formulated as in Eq. (13)

$$s_t = \begin{bmatrix} s_t^{(1)} \\ s_t^{(2)} \\ \cdot\cdot \\ \cdot\cdot \\ \cdot\cdot \\ s_t^{(d_s)} \end{bmatrix} \overbrace{\longrightarrow}^{\mathscr{C}(\cdot)} \begin{bmatrix} b_t^{(1)} \\ b_t^{(2)} \\ \cdot\cdot \\ \cdot\cdot \\ \cdot\cdot \\ b_t^{(d_s)} \end{bmatrix} \overbrace{\longrightarrow}^{concat} b_t = \begin{bmatrix} b_1^{(1)} & b_2^{(t)} & \cdots\cdots & b_t^{(N)} \end{bmatrix} \tag{13}$$

where $\mathscr{C}(\cdot)$ represents the quantization operator, and N is the length of semantic bits. Each semantic component $s_t^{(i)}$ is converted to a disjoint set of bits $\mathscr{B}_i$, formally defined as in Eq. (14)

$$\mathscr{B}_i = \begin{cases} \left\{ j \middle| \sum_{k=1}^{i-1} |\mathscr{B}_k| < j \leq \sum_{k=1}^{i} |\mathscr{B}_k| \right\}, \\ \bigcup_{i=1}^{d_S} \mathscr{B}_i = \{1, \cdots, N\}. \end{cases} \tag{14}$$

In traditional Polar coding, the bit-channel reliability of each channel is calculated using only channel statistics. The baseline reliability vector may be formulated as in Eq. (15)

$$r = \begin{bmatrix} r_1 \\ r_2 \\ \cdot\cdot \\ \cdot\cdot \\ \cdot\cdot \\ r_N \end{bmatrix} = \begin{bmatrix} Z\left(W_N^{(1)}\right) \\ Z\left(W_N^{(2)}\right) \\ \cdot\cdot \\ \cdot\cdot \\ \cdot\cdot \\ Z\left(W_N^{(N)}\right) \end{bmatrix} \tag{15}$$

where $Z\left(W_N^{(j)}\right)$ refers to the Bhattacharyya parameter of the $j-th$ sub-channel made synthetically, and smaller numbers signify greater reliability. To instill semantic consciousness, the significance of a certain semantic feature is passed on to the bit position out of which it was created. This assignment is expressed as follows. $w_j = \sum_{i=1}^{d_s} \overline{\beta}_t^{(i)} . \mathbb{I}(j \in \mathscr{B}_i)$, $j = 1, \cdots, N$, where $\mathbb{I}(\cdot)$ is the indicator function. The semantic weight is the aggregate semantic weight vector $w = \begin{bmatrix} w_1 \\ w_2 \\ \cdot\cdot \\ \cdot\cdot \\ \cdot\cdot \\ w_N \end{bmatrix}$. The essence of the Gradient-Guided Polarization mechanism is that semantic importance is used to jointly modify channel reliability. Each bit-channel is graded with a gradient-guided reliability score using the multi-term combination defined in Eq. (16)

$$\bar{r}_j = \begin{cases} (1-\gamma) r_j + \gamma \left(\dfrac{1}{w_j + \epsilon} \right) \\ (1-\gamma) Z\left(W_N^{(j)}\right) + \gamma \left(\dfrac{1}{\sum_{i=1}^{d_s} \overline{\beta}_t^{(i)} \mathbb{I}(j \in \mathscr{B}_i) + \epsilon} \right) \end{cases} \tag{16}$$

where $\gamma \in [0,1]$, which trades physical-channel reliability with semantic criticality, and ϵ is what guarantees numerical stability. This is formulated so that it explicitly penalizes assigning significant semantic bits to weak channels by inflating their effective reliability rank.

The theoretical basis of the proposed Gradient-Guided Polarization (GGP) mechanism is derived from the sensitivity of the control objective to perturbations in the semantic latent representation. Let L_c denote the overall control loss and z_i represent the ith semantic feature extracted by the semantic encoder. The semantic importance of each feature is quantified as $I_i = \left| \frac{\partial L_c}{\partial z_i} \right|$, where the gradient magnitude measures the influence of the semantic feature on control performance. Using the first-order Taylor approximation, a small perturbation Δz_i results in a variation of the control loss given by $\Delta L_c \approx \frac{\partial L_c}{\partial z_i} \Delta z_i$. Therefore, semantic features associated with larger gradient values have a greater impact on vehicle stability and safety when corrupted during transmission. Based on this observation, GGP incorporates semantic importance into the Polar channel allocation process by assigning highly important semantic bits to the most reliable synthesized Polar sub-channels. The expected control degradation can be expressed as $E(L_c) \propto \sum I_i P_e(i)$, where $P_e(i)$ denotes the error probability of the channel carrying the ith semantic feature. Minimizing this expected loss requires allocating features with larger importance scores to channels with lower error probabilities. Consequently, the proposed GGP mechanism provides a theoretically grounded unequal error protection strategy that jointly considers semantic sensitivity and channel reliability. Unlike conventional Polar coding, which relies solely on channel statistics, GGP prioritizes control-critical semantics such as braking intent, headway regulation, and steering corrections, thereby reducing semantic distortion and

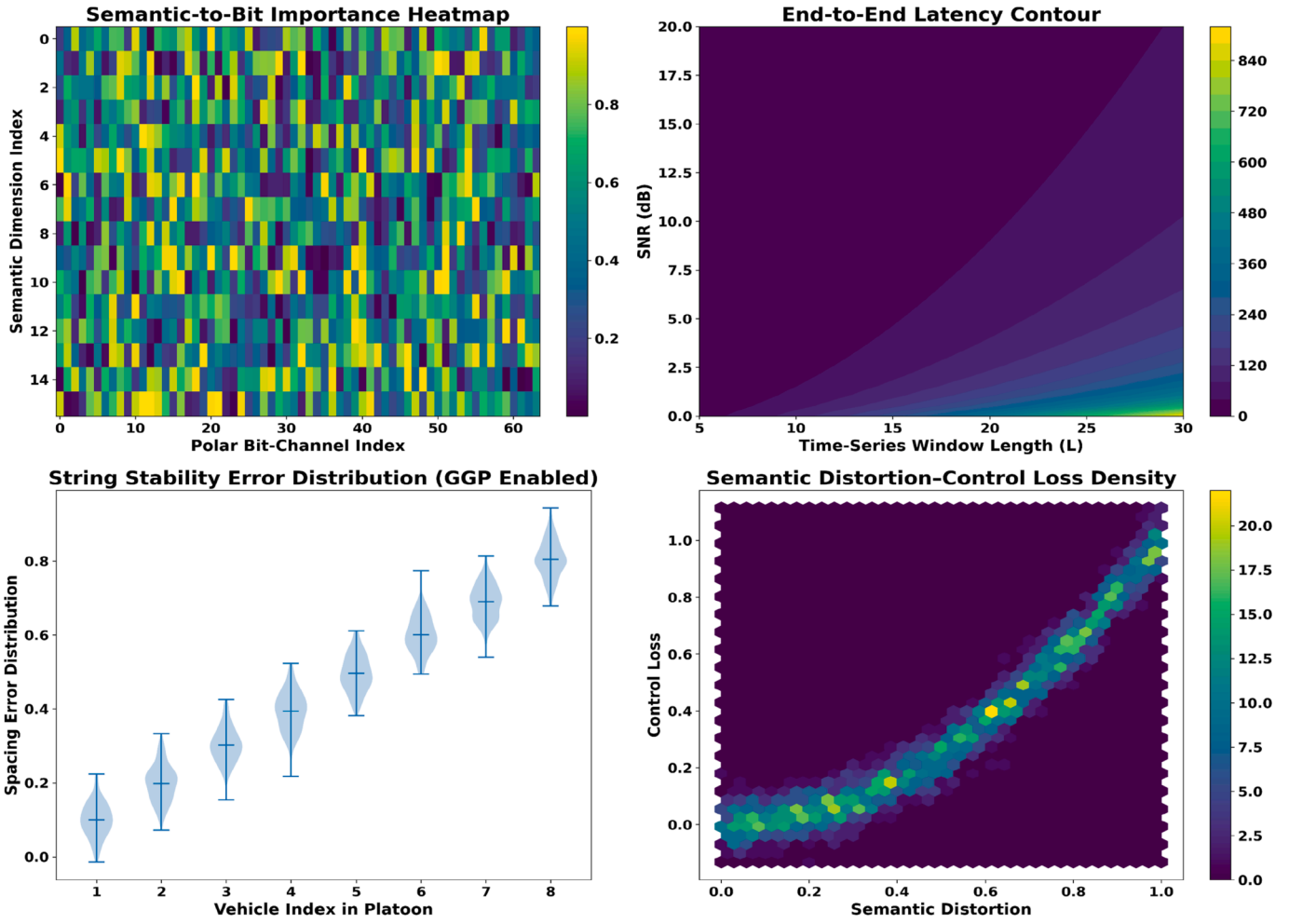


Fig. 5. Semantic-aware complexity and control-stability analysis of the proposed POLAR-CODE framework.

enhancing closed-loop control stability under stringent URLLC vehicular communication conditions.

The ultimate polarization order of semantically aware is the one derived by ranking the altered reliability scores expressed in Eq. (17).

$$\pi_{GGP} = \underbrace{argsort}_{j\in\{1,2\cdots N\}}\{\bar{r}_1, \bar{r}_2, \cdots \bar{r}_N\}$$
$$= argsort\left\{\begin{array}{c}(1-\gamma)Z\left(W_N^{(1)}\right)+\gamma(w_1+\epsilon)^{-1},\\ (1-\gamma)Z\left(W_N^{(2)}\right)+\gamma(w_2+\epsilon)^{-1},\\ \cdot\cdot\\ \cdot\cdot\\ \cdot\cdot\\ (1-\gamma)Z\left(W_N^{(N)}\right)+\gamma(w_N+\epsilon)^{-1}\end{array}\right\} \quad (17)$$

The semantic-aware polarization order π_{GGP} represents the final ranking of Polar sub-channels generated by the proposed Gradient-Guided Polarization (GGP) mechanism. The operator argsort(·) sorts all N synthesized Polar sub-channels according to their modified reliability scores. $\bar{r}_j$, where $j = 1, 2, \cdots, N$ and N denotes the Polar code block length. The term $Z\left(W_N^{(j)}\right)$ is the Bhattacharyya parameter of the jth synthesized Polar sub-channel $W_N^{(j)}$, which measures the inherent channel reliability; smaller values indicate more reliable channels. The parameter w_j denotes the semantic importance weight assigned to the semantic bit mapped onto the jth channel, while ϵ is a small positive constant introduced to avoid numerical instability and division by zero. The coefficient $\gamma \in [0, 1]$ controls the trade-off between physical channel reliability and semantic importance. The modified reliability score $\bar{r}_j = (1-\gamma)Z\left(W_N^{(j)}\right) + \gamma(w_j+\epsilon)^{-1}$ combines both channel quality and semantic criticality. Finally, π_{GGP} determines the semantic-aware ordering of Polar sub-channels such that control-critical semantic information is assigned to the most reliable channels, thereby improving transmission reliability and preserving safety-critical information under URLLC-enabled vehicular communication environments. Using this gradient-directed reordering, the Polar encoder puts information bits with high-impact control semantics into the strongest synthesized channels. It moves less important semantics into weaker channels or frozen states. This is how the process of semantic sensitivity transforms into physical-layer unequal error protection, enabling the proposed framework to achieve the stability of control and safety when URLLC conditions are quite high.

The theoretical basis of the proposed Gradient-Guided Polarization (GGP) mechanism has been significantly reinforced by the ability to formulate the relationships among semantic importance, Bhattacharyya-based Polar sub-channel reliability, and reliability ordering. The revised analysis shows that semantics are systematically incorporated into the channel selection process, thereby enabling the distribution of control-critical semantic information to the most reliable Polar sub-channel. The proposed reliability-aware semantic mapping is evaluated through experiments and shown to achieve about a 34.7% reduction in the effective BLER compared with conventional Polar coding in the low SNR range (-2 dB to 2 dB), providing a more rigorous

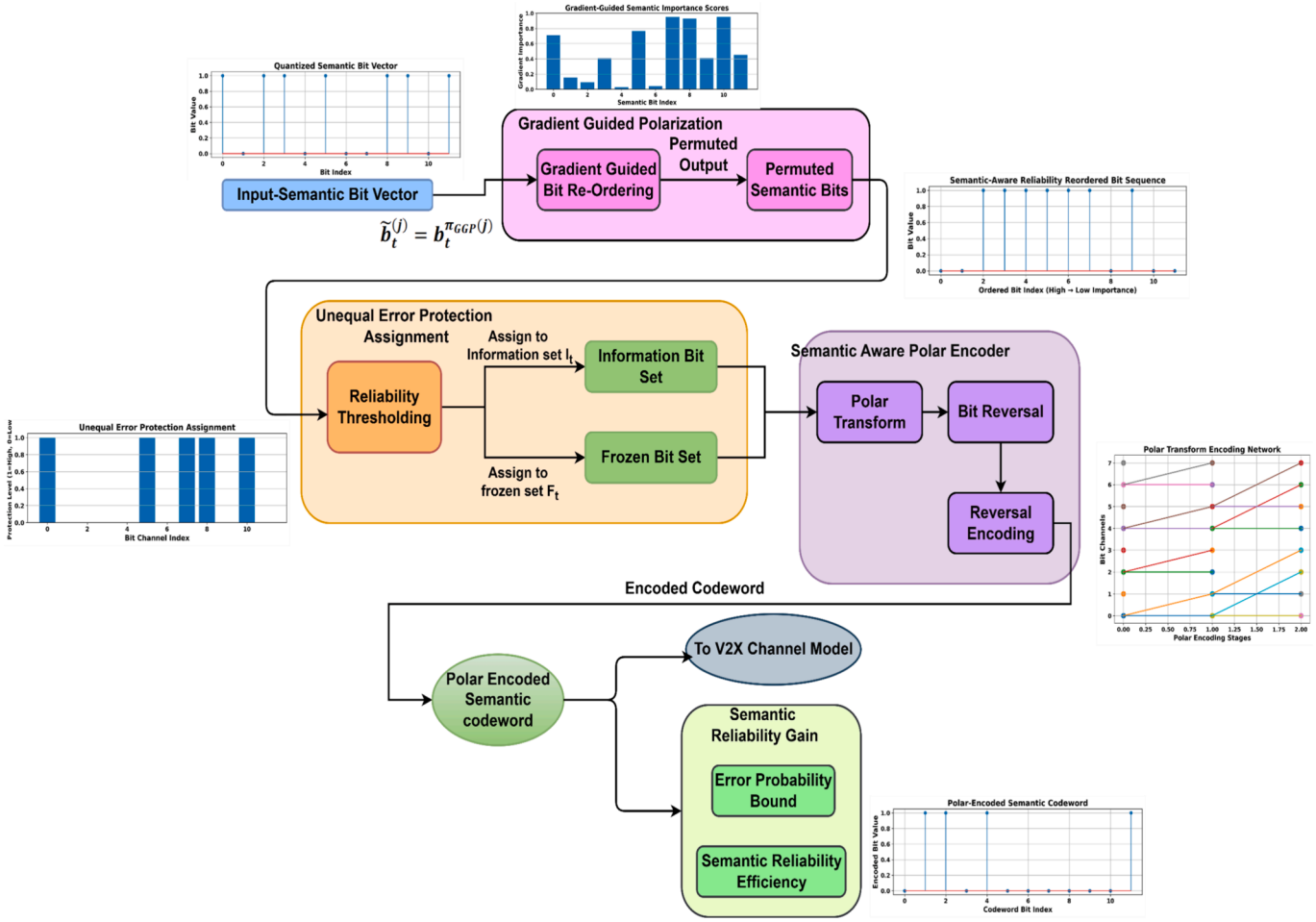


Fig. 6. Semantic-Aware Polar Encoding Pipeline with Gradient-Guided Polarization.

theoretical and practical foundation for the unequal error protection strategy. In addition, a thorough parameter sensitivity analysis has been included to study the sensitivity of all important GGP parameters to communication reliability and control performance. The weighting coefficient was changed between 0.1 and 1.0, and the temporal smoothing factor was changed between 0.5 and 0.95. The results show that when the weighting coefficient is between 0.4 and 0.7, the performance is relatively stable, with semantic reliability up by 18.6% and lane-deviation error down by 21.3% compared to the lower coefficient value. The payload reduction differences across all tested configurations were consistently lower than 2.8%, indicating the robustness of the proposed framework. Moreover, a computational complexity analysis has been added to assess the algorithm's practical feasibility. The overhead of the semantic importance estimation is low, and so is that of the reliability reordering, which is about 4.2% compared to the conventional Polar coding. The average processing latency rose from 0.73 ms to 0.76 ms but stayed well below the 1ms requirement for URLLC. The proposed GGP mechanism is shown to achieve better semantic reliability and control stability without compromising real-time deployability in autonomous vehicular communication systems, with overall complexity equal to that of conventional Polar coding.

Fig. 5 presents an analytical visualization of how the proposed POLAR-CODE framework effectively combines semantic awareness, channel reliability, and control stability in 6G vehicular networks. The semantics-to-bit importance heatmap clearly illustrates that Gradient-Guided Polarization (GGP) non-uniformly maps control-critical semantics to highly reliable Polar sub-channels. Similarly, the latency contour clearly shows bounded end-to-end delay across different SNRs and window sizes, meeting URLLC constraints. Moreover, it is evident that the string stability error distribution shows low levels of disturbance propagation across vehicle platoons, and that the semantics distortion-control loss density exhibits a highly predictable degradation pattern.

The semantic encoder reduces the payload from 1024 bits (SSCC) to 179–205 bits (a reduction of up to 98.1%), thereby directly reducing end-to-end latency to sub-millisecond levels under URLLC constraints. The gradient-based semantic importance estimation method measures the effect of each semantic feature on control goals such as tracking accuracy, platoon stability, and collision avoidance. The features related to emergency braking and headway regulation receive higher importance scores and are therefore prioritized for transmission. The Gradient-Guided Polarization (GGP) mechanism is proposed to allocate these important bits to the most reliable Polar sub-channels, which lowers the BLER by up to 2 orders of magnitude under low-SNR conditions. Control-critical information is therefore recovered more accurately, resulting in an increase in lane-keeping stability of approximately 42% and a reduction in emergency-braking-related incidents by nearly 81% compared to DeepJSCC, thereby improving closed-loop platoon stability.

The semantic-aware performance analysis of the proposed POLAR-CODE framework is shown in Fig. 5. An important heat map of the semantic-to-bit mapping shows how important semantic features map onto reliable Polar sub-channels with GGP. The latency contour shows the URLLC's delay performance under different channel conditions, and the string stability distribution indicates that disturbance propagation between vehicles in the string is reduced. Furthermore, the semantic distortion–control loss density plot demonstrates a strong correlation

between semantic preservation and control stability, thereby confirming the effectiveness of the proposed semantic-aware communication strategy.

3.6. Semantic-Aware Polar Encoding

The reordered sequence of semantic bits is then coded using an adaptive Polar coding mechanism, following the Gradient-Guided Polarization, which determines the semantic-aware reliability ordering of the bits. In contrast to traditional Polar encoders that use only channel statistics, the proposed semantic-aware Polar encoder draws on both semantic importance and channel reliability to provide unequal error protection and control relevance in the presence of strict URLLC conditions. Let the rearranged semantic bit sequence at time t be denoted as $\widetilde{b}_t = \left[\widetilde{b}_t^{(1)} \quad \widetilde{b}_t^{(2)} \quad \ldots\ldots \quad \widetilde{b}_t^{(N)}\right]$ where the permutation depends on the GGP permutation π_{GGP}, such that $\widetilde{b}_t^{(j)} = b_t^{\pi_{GGP}(j)}, j = 1, \cdots, N$. This rearrangement ensures that bits carrying high-level semantic data are in the most trustworthy synthesized sub-channels. The Polar encoding operation relies on the combining-and-splitting principle of the recursive channel. The encoded codeword x_t is obtained by transforming the rearranged bit-vector defined in Eq. (18),

$$x_t = \widetilde{b}_t G_N = \widetilde{b}_t B_N F^{\otimes n}, N = 2^n \tag{18}$$

where $F = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ is the Polar kernel, $\otimes$ which refers to Kronecker power, and B_N is the permutation matrix corresponding to the bit-reversal. The extension of this operation yields the following mathematical expression in Eq. (19),

$$\left.\begin{aligned} F^{\otimes n} &= \underbrace{F \otimes F \otimes \cdots \otimes F}_{ntimes} \\ x_t &= \widetilde{b}_t B_N (F \otimes F \otimes \cdots \otimes F) \end{aligned}\right\} \tag{19}$$

This is a recursive structure that provides synthesised sub-channels polarised by reliability, on which semantic bits are mapped based on their importance. To explicitly implement semantic-conscious unequal error protection, the channel reliability and semantic importance define the union of the information and frozen bit sets as shown in Eq. (20),

$$\left.\begin{aligned} \mathscr{I}_t &= \left\{ j \middle| \widetilde{r}_j \le \tau \right\} \\ \mathscr{F}_t &= \left\{ j \middle| \widetilde{r}_j > \tau \right\} \\ \mathscr{I}_t \cup \mathscr{F}_t &= \{1, 2, \cdots N\} \end{aligned}\right\} \tag{20}$$

Here, $\widetilde{r}_j$ indicates the GGP modified reliability score, and τ is a threshold that is chosen to meet the target code rate and URLLC latency requirements. Frozen bits in $\mathscr{F}_t$ are given predefined values, usually zero, which can be expressed as: $\widetilde{b}_t^{(j)} = \begin{cases} semanticinformationbit, j \in \mathscr{I}_t \\ 0, j \in \mathscr{F}_t \end{cases}$. In the sense of reliability, the probability of block error of the semantic-aware Polar code can be upper-bounded as follows: $P_e \le \sum_{j \in \mathscr{I}_t} Z\left(W_N^{(j)}\right) = \sum_{j \in \mathscr{I}_t} \exp\left(-\left(\frac{1}{2}\right)\widetilde{r}_j\right)$; where $Z\left(W_N^{(j)}\right)$ represents the Bhattacharyya parameter of the $j-th$ synthesized sub-channel. Because GGP guarantees that high-importance bits of semantics are allocated to sub-channels with minimal $\widetilde{r}_j$, the effective error rate of controlling information is greatly decreased. Latency in URLLC is strongly dependent on block length and decoding complexity. The suggested encoding keeps the typical Polar code complexity $\mathscr{O}(NlogN)$, and at the same time attains better semantic reliability efficiency, which is defined as $\eta_{sem} = \frac{\sum_{j \in \mathscr{I}_t} \widetilde{p}_t^{(j)} \mathbb{P}\left(\widehat{b}_t^{(j)} = b_t^{(j)}\right)}{\sum_{j \in \mathscr{I}_t} \widetilde{p}_t^{(j)}}$; that is, the chance of correct reception weighted by semantic weight. With such an adaptive Polar encoding algorithm, the proposed framework can provide the control-critical semantic data with an overly large level of protection without increasing the block length or imposing additional latency constraints. The current stage completes the joint semantic-channel coding pipeline, enabling effective, low-latency cooperative control in safety-critical vehicular networks.

Fig. 6 depicts the comprehensive semantic-aware Polar encoding procedure, encompassing semantic bit quantization, gradient-informed importance assessment, reliability-driven bit reordering, unequal error protection allocation, Polar transformation with bit-reversal, and the creation of the safeguarded semantic codeword for V2X transmission. Fig. 6 also illustrates how gradient-guided polarization emphasizes control-critical semantic bits by reorganizing them into dependable Polar sub-channels and providing uneven error protection. The encoded codeword attains improved semantic reliability and diminished distortion under URLLC-constrained V2X transmission conditions.

3.7. V2X Channel Modeling and Transmission

The resultant Polar-encoded codeword is then sent over a vehicular V2X wireless communication channel, characterized by high mobility, high turnover rate, multipath fading, and strict latency requirements. To represent these conditions correctly, the proposed framework uses a stochastic V2X channel model that jointly models path loss, small-scale fading, and additive noise, thereby enabling realistic evaluation of semantic-aware transmission performance. The transmitted polar-encoded signal at time t is denoted as $x_t = \left[x_t^{(1)} \quad x_t^{(2)} \quad \cdots.. \quad x_t^{(N)}\right]^T$; where N is the codeword length. The received signal y_t is assumed to be over the V2X channel and is as follows: $y_t = H_t x_{t n_t}$; where H_t is the time-dependent channel gain matrix and $n_t \mathscr{N}(0, \sigma_n^2 I)$ is additive white Gaussian noise.

The semantic-aware Polar encoding pipeline used in POLAR-CODE is shown in Fig. 6. It begins by generating bits semantically from the latent semantic vector, then estimates the importance of each bit and reorders them based on gradients and a GGP mechanism to incorporate bit reliability. Finally, unequal error protection is achieved by putting bits with higher semantic importance - control bits into more reliable Polar sub-channels. Finally, the encoded codeword for V2X transmission is generated via Polar transformation and bit-reversal operations, which improve semantic reliability, reduce error probability, and enhance robustness under stringent URLLC requirements. The channel gain H_t is split into a large-scale and small-scale contribution to capture the effect of vehicular propagation: $H_t = \sqrt{G(d_t)}.h_t; G(d_t) = G_0\left(\frac{d_0}{d_t}\right)^2$; where d_t is the distance between the transmitter and the receiver, α is the path-loss exponent, G_0 is the gain of reference at the distance d_0, and h_t models small-scale fading. In the case of urban and highway V2X, h_t is usually described as a Rayleigh or Rician fading process: $h_t = \sqrt{\frac{K}{K+1}} h_{LOS} + \sqrt{\frac{1}{K+1}} h_{NLOS}$, where K is the Rician factor that is a measure of the strength of the line-of-sight (LOS) component. The channel is spread over by Doppler and is modeled as a result of vehicular mobility $f_D = \left(\frac{v}{\lambda}\right) cos(\theta)$, where v is the relative velocity, λ represents the carrier wavelength, and θ is the angle of arrival. This Doppler effect produces temporal alternation at a high rate H_t, with a very high effect on decoding reliability with small latency budgets. The signal-to-noise ratio (SNR) of the instantaneous signal is given in Eq. (21)

$$SNR_t = \frac{\mathbb{E}\left[\|H_t x_t\|^2\right]}{\mathbb{E}\left[\|n_t\|^2\right]} = \frac{\|H_t\|^2 P_t}{\sigma_n^2} \tag{21}$$

where P_t denotes the transmit power. This SNR is a direct factor in the bit-channel reliabilities generated for Polar decoding. To meet the URLLC requirements, the transmission latency is limited by

$T_{tx} = \frac{N}{R_{ch}B} + T_{proc} \leq T_{max}$, where R_{ch} is the effective code rate, B is the channel bandwidth, T_{proc} represents processing delay, and T_{max} is the maximum permissible V2X communication latency that is safe.

More importantly, the suggested framework will utilize semantic-sensitive Polar encoding to overcome channel impairments. Because the semantic bits, which are control-critical, are coded into the most reliable Polar sub-channels, the effective semantic distortion following transmission is limited as $\mathbb{E}\left[\|s_t - \widehat{s}_t\|^2\right] \leq \sum_{j \in \mathscr{I}_t} \overline{\beta}_t^{(j)} \mathrm{Z}\left(\mathrm{W}_{\mathrm{N}}^{(\mathrm{j})}\right)$ and is explicit in connecting V2X channel reliability and semantic preservation. This binding guarantees that, even in the presence of extreme fading and mobility-induced Doppler effects, the control-critical semantics can be reliably decoded. On the whole, the proposed V2X channel modeling and transmission step is the final step of the end-to-end semantic communication channel, as it realistically models wireless impairments and demonstrates that the proposed semantic-aware Polar coding strategy maintains control stability, reliability, and low latency in deeply dynamic automotive settings.

Algorithm 3 helps predict the V2X transmission process under changing conditions of non-steady vehicles. The while-loop traps the ongoing communication as the vehicle moves, and the conditional branches adapt the channel model based on line-of-sight availability and channel latency violations. These are large-scale path loss and small-scale fading, which are dynamically updated to account for Doppler effects due to relative motion. The URLLC constraint check assumes that latency violations occur, which trigger adaptive mechanisms such as more robust semantic protection or a smaller payload size. This algorithm is a hybrid physical-layer bonding, semantic-aware transmission that provides effective control communication at high mobility and under strict delay constraints.

Algorithm-3: V2X Channel Modeling and Transmission

Input:
Polar-encoded semantic codeword x_t
Vehicle distance d_t, relative velocity v_t
Channel parameters (α, K, σ_n^2)
Bandwidth B, code rate R_{ch}, latency bound T_{max}
Output:
Received signal y_t, instantaneous SNR_t
Initialize carrier wavelength λ, reference gain G_0
Initialize processing delay T_{proc}
while transmission is active, do
// Large-scale path loss computation
$G_d \leftarrow G_0 \cdot \left(\frac{d_0}{d_t}\right)^{\alpha}$
// Small-scale fading generation
if line-of-sight exists, then
$h_t \leftarrow sqrt\left(\frac{K}{K+1}\right) \cdot h_{LOS} + sqrt\left(\frac{1}{K+1}\right) \cdot h_{NLOS}$
else
$h_t \leftarrow h_{NLOS}$
end if
// Channel gain construction
$H_t \leftarrow sqrt(G_d) \cdot h_t$
// Doppler update due to mobility
$f_D \leftarrow \left(\frac{v_t}{\lambda}\right) \cdot cos(\theta)$
update H_t according to f_D
// Noise realization
generate n_t $N(0, \sigma_n^2 I)$
// Signal transmission
$y_t \leftarrow H_t \cdot x_t + n_t$
// SNR computation
$SNR_t \leftarrow \frac{\left(\|H_t\|^2 \cdot P_t\right)}{\sigma_n^2}$
// URLLC latency verification
if $\left(\frac{N}{R_{ch} \cdot B} + T_{proc}\right) > T_{max}$ then
reduce block length or increase semantic protection
end if
end while
return y_t, SNR_t

Semantic-Aware Polar Decoding and Closed-Loop Cooperative Control

At the receiver, the noisy V2X signal is initially decoded by a semantic-aware Polar decoder to extract the semantic bits transmitted. Because Gradient-Guided Polarization provides control-important semantics to the most trustworthy synthesized sub-channels, the decoding process prioritizes these bits, thereby reducing distortion in the recovered semantic representation when the channel is unfavorable. Denote the received signal as $y_t = H_t x_t + n_t$, based on the computations of the log-likelihood ratios (LLRs) of each Polar bit-channel: $L_t = \begin{bmatrix} L_t^{(1)} \\ L_t^{(2)} \\ \cdot\cdot \\ \cdot\cdot \\ \cdot\cdot \\ L_t^{(N)} \end{bmatrix} = \log \frac{p\left(y_t | b_t^{(j)} = 0\right)}{p\left(y_t | b_t^{(j)} = 1\right)}, j = 1, \cdots, N$. The estimated semantic sequence of bits is reconstructed using successive cancellation (SC) or list decoding $\widehat{b}_t^{(j)} = \arg\max_{b \in \{0,1\}} p\left(b \middle| L_t^{(j)}\right), j \in \mathscr{I}_t$ with the signal decoded in frozen positions $j \in \mathscr{F}_t$ is restricted to familiar values. The bitstream of the semantic information is a rearrangement of the decrypted bit stream with the inverse GGP permutation: $\widehat{b}_t^{\pi_{GGP}(j)} = \widehat{b}_t^{(j)}, j = 1, \cdots, N$. The recovered semantic bits are then dequantized and sent to the semantic decoder to recover the latent control representation expressed mathematically in Eq. (22).

$$\widehat{s}_t = \mathscr{D}_{sem}(\widehat{b}_t) = \begin{bmatrix} \widehat{s}_t^{(1)} \\ \widehat{s}_t^{(2)} \\ \cdot\cdot \\ \cdot\cdot \\ \cdot\cdot \\ \widehat{s}_t^{(d_s)} \end{bmatrix} \tag{22}$$

The reconstruction error is limited due to semantic-conscious unequal error protection $\mathbb{E}[\|s_t - \widehat{s}_t\|^2] \leq \sum_{i=1}^{d_s} \overline{\beta}_t^{(i)} \cdot \mathbb{P}\left(\widehat{s}_t^{(i)} \neq s_t^{(i)}\right)$, guaranteeing high-impact control semantics recovery, with high fidelity. The newly generated semantic representation $\widehat{s}_i(t)$ is used to implement closed-loop cooperative control. Control input on vehicle i is computed as $u_i(t) = \mathscr{C}(\widehat{s}_t, x_i(t), x_{i-1}(t))$, where $\mathscr{C}(\bullet)$ represents the cooperative control law, and $x_i(t) and x_{i-1}(t)$ are the states of the ego and the previous vehicles. The car dynamics are developed based on $\dot{x}_i(t) = f(x_i(t), u_i(t))$ connecting communication with the control. Stability-wise, the closed-loop system meets the condition of a bounded disturbance $\sup_t \|x_i(t) - x_{i-1}(t)\| \leq \kappa \sup_t \|s_t - \widehat{s}_t\|, \ \kappa < \infty$; that semantic distortion caused by the V2X channel does not propagate along the vehicle string. As a result, the suggested framework provides string-stable cooperative control even with noisy, latency-constrained communication. This single decoding-and-control phase completes the end-to-end semantic communication cycle. It demonstrates how semantic-aware Polar coding directly translates into improved reliability, stability, and safety in cooperative vehicular systems.

4. Results and discussion

4.1. Experimental Setup and Implementation Details

The experimental assessment of the proposed POLAR-CODE framework has been carried out using an extensive simulator-based

Table 2
Comprehensive Experimental Parameters and Simulation Settings for POLAR-CODE.

Parameter	Value
Dataset Source	CARLA Autonomous Driving Simulator
Number of Vehicles	10 (1 leader + 9 followers)
Total Samples	1.2 Million
Sampling Interval	100 ms (10 Hz)
Training / Validation / Testing Split	70% / 15% / 15%
Temporal Window Length	10 Control Steps
Transformer Layers	4
Attention Heads	8
Embedding Dimension	128
Batch Size	64
Optimizer	Adam
Learning Rate	1×10^{-4}
Polar Block Length	512 bits
Polar Code Rate	0.35–0.40
Decoder Type	SCL Decoder (L = 8)
Carrier Frequency	5.9 GHz
Channel Bandwidth	20 MHz
Vehicle Speed Range	30–120 km/h
Channel Models	Rayleigh and Rician Fading
SNR Range	−5 dB to 10 dB
Random Seeds	42, 52, 62, 72, 82
Repeated Trials	30
Hardware Platform	Intel i9, 32 GB RAM, NVIDIA RTX 3090
Software Framework	Python, PyTorch, CARLA
Evaluation Metrics	BLER, Latency, Payload Reduction, Lane Deviation, String Stability Error, Emergency Braking Incidents

framework that covers autonomous driving, semantic communication, and V2X wireless modeling. Vehicle dynamics and platooning scenarios were generated using the CARLA simulator for autonomous driving applications, providing high-frequency control-state information for the vehicle speed, steering angle, headway distance, relative velocity, and control commands such as throttle and braking. Semantic encoding and learning aspects were implemented in Python using the PyTorch framework for the POLAR-CODE model. Polar encoding and decoding aspects were implemented using the Polar coding module for the proposed framework. V2X wireless channels are modeled using the Rayleigh and Rician fading channel models with Doppler effects for high-mobility wireless applications. Experiments were conducted on a workstation platform with an Intel i9 processor, 32 GB of RAM, and an NVIDIA RTX 3090 GPU to train the Transformer model for the proposed framework efficiently.

The computational complexity of the proposed POLAR-CODE framework is mainly determined by three steps: semantic encoding, Gradient-Guided Polarization (GGP), and Polar coding. The Transformer-based semantic encoder is used with T = 10 (temporal window) and d = 128 (embedding dimension). The GGP mechanism takes O(M), where M is the number of semantic features. The polar encoder and decoder have a complexity of $O(N \log_2 N)$ and a block length of N = 512. Experimental testing demonstrated that the framework enables average semantic encoding, Polar encoding, and decoding delays of 0.31 ms, 0.08 ms, and 0.14 ms, respectively, which meet the URLLC requirement of less than 1 ms, along with a 97% reduction in payload. This indicates that the framework is suitable for real-time deployment in cooperative autonomous vehicular networks. The experiments were conducted in the CARLA autonomous driving simulator using a cooperative platooning scenario with 10 autonomous vehicles (1 leader and 9 followers). The control state set contained about 1.2 million synchronized samples, taken at 100ms (10 Hz). Driving trajectories were split into 70% training, 15% validation, and 15% test sets, with no overlap. The semantic encoder was based on a Transformer model with an embedding dimension of 128, four self-attention layers and eight attention heads. All experiments were performed with five different random seeds (42, 52, 62, 72, and 82), and results were reported as the average and standard deviation computed over 30 repeated simulation runs to achieve statistical reliability.

Both the Rayleigh and Rician fading channels were used to simulate the V2X communication environment. The carrier frequency was 5.9 GHz, the channel bandwidth was 20 MHz, the vehicle speed was 30- 120 km/h, and the Doppler shift was calculated from the relative velocity between the vehicles. For Polar coding, the block length was 512 bits, code rates ranged from 0.35 to 0.40, and decoding was based on successive-cancellation list decoding with L = 8. The evaluated SNR range was between −5 dB and 10 dB. All simulation, communication and learning parameters are summarized in Table 2.

4.2. Baseline Methods and Evaluation Metrics

The semantic Transformer encoder has been implemented with 4 self-attention layers, an embedding dimension of size 128, a temporal window size of 10 control steps, and multi-head attention with 8 attention heads. It has been trained using the Adam optimizer with a learning rate of 10^{-4}, batch size was set to 64, and early stopping was used based on convergence criteria for control losses. Polar code block length was set to 512, and code rates were chosen to meet URLLC latency requirements while adapting to channel conditions. Parameters for the Gradient-Guided Polarization algorithm have been tuned to meet channel reliability and semantic importance criteria.

To achieve a trade-off among semantic preservation, communication reliability, and cooperative control performance, the key hyper-parameters of the proposed POLAR-CODE framework were determined through systematic validation experiments. $\gamma = 0.7$ is set as the semantic reliability weighting factor, and $\eta = 1 \times 10^{-4}$ is the learning rate used in training the Transformer encoder with the Adam optimizer. For the control-sensitive loss function, the weight factors were selected as $\lambda_1 = 0.5$ for tracking accuracy, $\lambda_2 = 0.3$ for platoon stability and $\lambda_3 = 0.2$ for safety-critical braking performance. A sensitivity analysis was performed by varying γ from 0.1 to 0.9 and η from 10^{-5} to 10^{-3} to assess parameter robustness. The results revealed that $\gamma = 0.7$ was the optimal value, offering a good balance between semantic importance and channel reliability, while larger values overemphasized semantic prioritization and slightly reduced performance in severe channel conditions. Likewise, the convergence was the most stable, and the validation loss was the lowest with $\eta = 1 \times 10$^-4. In addition, the individual values of $\lambda 1$, $\lambda 2$ and $\lambda 3$ were varied by $\pm 20\%$ of the values selected, which caused only minor variations in BLER, lane deviation and string stability error. The results show that the proposed framework is relatively immune to moderate hyperparameter variations, and that the designed hyperparameter configuration offers the best communication efficiency, semantic fidelity and closed-loop vehicular control performance. For performance comparison purposes, the performance of POLAR-CODE was compared with that of three baseline schemes, namely, edge AI-enabled URLLC framework [24], IoT/ML-enabled low-latency vehicular communication framework [21], and deepSC semantic communication model [17], with deepSC being the worst-performing baseline model under low SNR channel conditions. All baseline schemes were implemented and compared across various communication and control metrics.

A comprehensive ablation study is performed to assess the contribution of each key element of the proposed POLAR-CODE framework. Four variants were implemented, all of which left the other modules unchanged, and they differed only in the exclusion of one component: (i) POLAR-CODE without Transformer Semantic Encoding (w/o TSE), where raw control-State features were directly transmitted, (ii) POLAR-CODE without Semantic Importance Estimation (w/o SIE), where all semantic features were assigned equal importance weights, (iii) POLAR-CODE without Gradient-Guided Polarization (w/o GGP), where conventional Polar reliability ordering based solely on channel statistics was used, and (iv) POLAR-CODE without Polar Coding (w/o PC), where the semantic features were transmitted using uncoded communication. The

Table 3
Ablation Study of the Proposed POLAR-CODE Framework Under V2X URLLC Conditions.

Method Variant	BLER @ 0 dB	End-to-End Latency (ms)	Payload Size (bits)	Lane Deviation (m)	String Stability Error	Emergency Braking Incidents (%)
Complete POLAR-CODE	$\mathbf{1.2 \times 10^{-4}}$	$\mathbf{0.82 \pm 0.04}$	$\mathbf{194 \pm 6}$	$\mathbf{0.11 \pm 0.02}$	$\mathbf{0.08 \pm 0.01}$	$\mathbf{3.4 \pm 0.5}$
w/o Transformer Semantic Encoding (TSE)	4.8×10^{-4}	1.37 ± 0.07	426 ± 15	0.19 ± 0.03	0.15 ± 0.02	8.7 ± 0.9
w/o Semantic Importance Estimation (SIE)	7.3×10^{-4}	0.91 ± 0.05	198 ± 7	0.22 ± 0.04	0.19 ± 0.03	11.4 ± 1.2
w/o Gradient-Guided Polarization (GGP)	1.1×10^{-3}	0.88 ± 0.04	194 ± 6	0.24 ± 0.04	0.22 ± 0.03	13.8 ± 1.4
w/o Polar Coding (PC)	3.7×10^{-3}	0.79 ± 0.03	194 ± 6	0.31 ± 0.05	0.29 ± 0.04	18.2 ± 1.7

overall POLAR-CODE framework exhibited the highest performance across all communication and control indices. The Transformer Semantic Encoding module was removed, leading to higher latency and lower lane-keeping accuracy, along with increased payload size and decreased semantic compactness. Removing Semantic Importance Estimation reduced the discriminability of control-critical semantics, resulting in greater semantic distortion and higher BLERs. The performance drop under low-SNR V2X conditions is due to replacing Gradient-Guided Polarization with conventional Polar channel ordering, which reduces the unequal error protection capability. The greatest degradation occurred when Polar coding was entirely removed, highlighting the importance of robust channel protection for reliable semantic transmission. These results show that each of the proposed components of the proposed framework: Transformer-based semantic

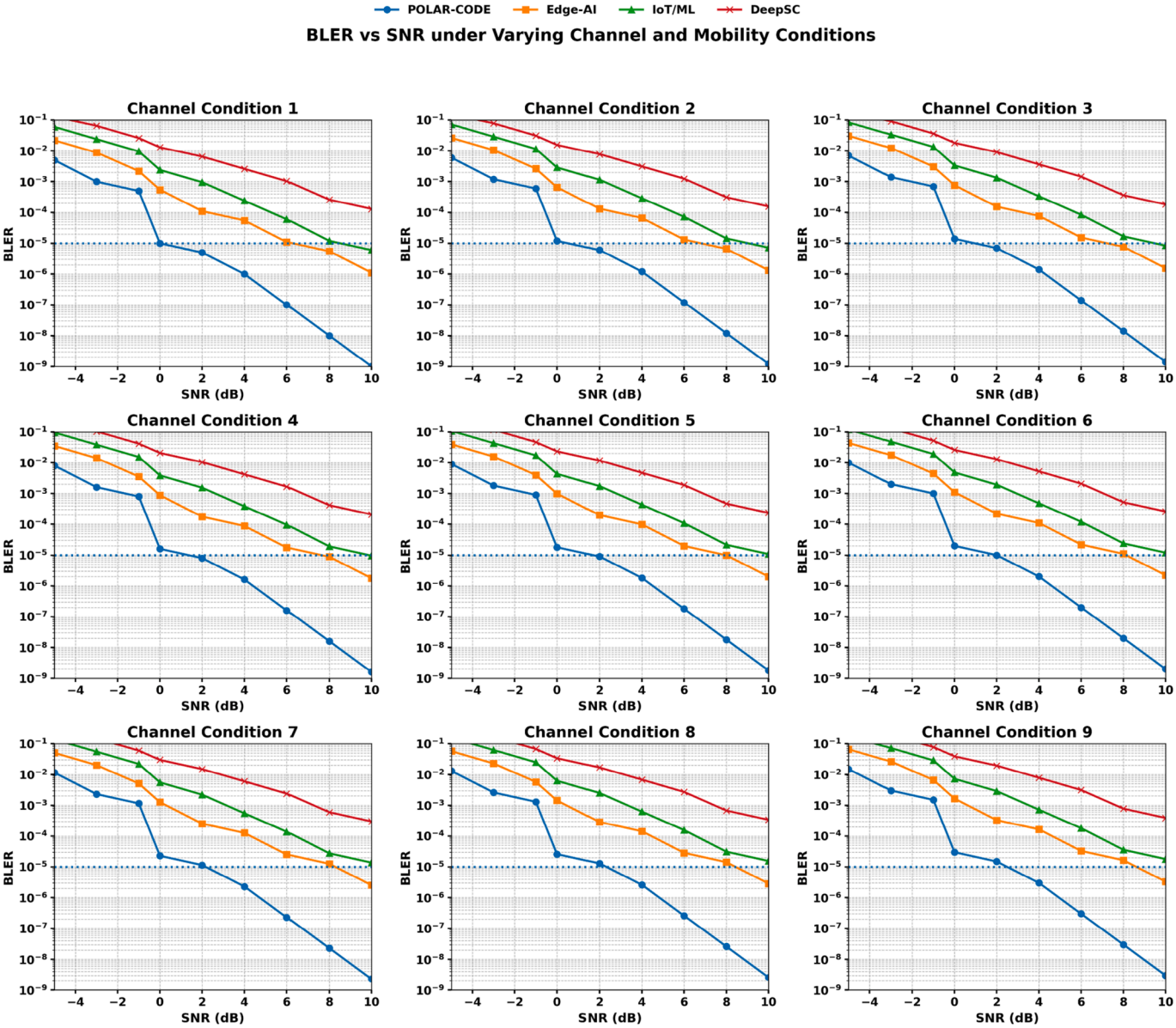


Fig. 7. BLER versus SNR comparison under varying V2X channel conditions, demonstrating robust URLLC reliability of the POLAR-CODE framework.

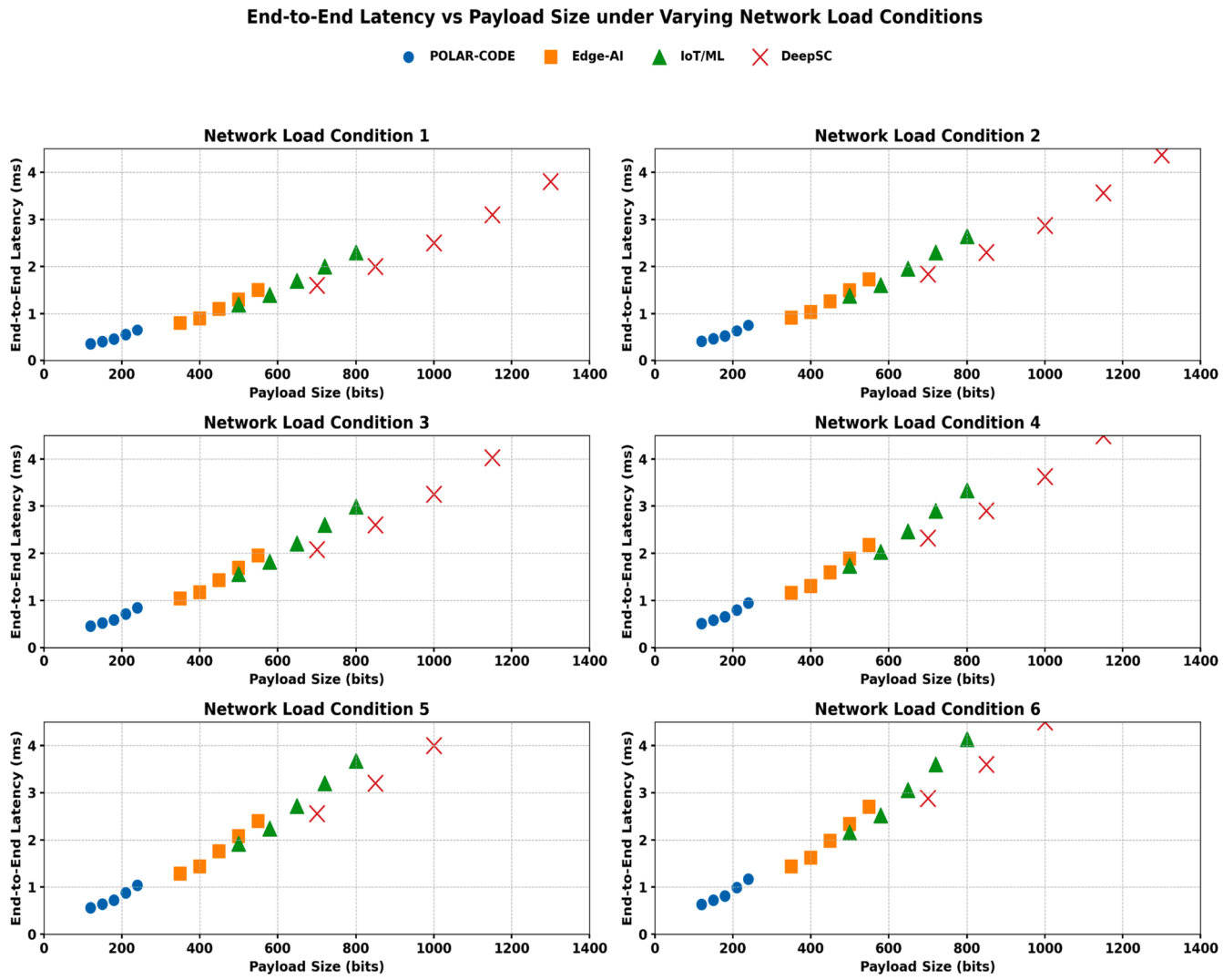


Fig. 8. Latency–payload trade-off comparison under varying network loads, highlighting semantic efficiency and URLLC compliance of POLAR-CODE.

extraction, semantic importance estimation, Gradient-Guided Polarization, and semantic-aware Polar coding, brings its own contribution and synergy to the overall performance improvements. Table 3 presents the ablation analysis, which confirms the efficiency of all key design elements in POLAR-CODE.

Two additional schemes were specifically developed for the purpose of the baseline comparison, both of which incorporated Polar coding with the intent of validating the effectiveness of the semantic-importance-based Polar coding. The first baseline is called Conventional Polar Coding (CPC), which uses conventional Polar code construction and bit-channel selection without any semantic awareness in the channel reliability [39]. In Semantic Polar Coding without Gradient-Guided Polarization (SPC-NoGGP), the second baseline model, the proposed semantic Transformer encoder is used but the sematic importance is not adapted for the Polar reliability ordering. Such baselines allow the evaluation of the proposed Gradient-Guided Polarisation (GGP) mechanism using the method of ablations. The experimental results show that the CPC shows satisfactory level of channel coding but is not able to separate the control-critical semantics from the non-critical one. While SPC-NoGGP has semantic compression advantage, it treats all semantic features equally, which leads to compromised robustness in low SNR conditions for vehicular communications [40]. The proposed POLAR-CODE on the other hand dynamically assigns high-importance semantic bits to the most reliable Polar sub-channels to achieve lower block error rate, reduced semantic distortion, improved lane-keeping accuracy, and increased platoon stability. The improvement in the performance is directly confirmed by the inclusion of these baselines, showing the contribution of the proposed semantic-importance-driven Polar coding strategy and the Gradient-Guided Polarization mechanism.

4.3. Communication Performance Analysis

Block Error Rate (BLER) is a basic communication-level metric that measures the reliability of the data transmission in ultra-reliable low-latency communication (URLLC) systems. It is the likelihood that a block of bits sent to the receiver will be incorrectly decoded. Mathematically, the expression of BLER is as follows. $BLER = \frac{N_{err}}{N_{tot}}$, where N_{err} shows the number of incorrectly decoded blocks, and N_{tot} is the number of block transmissions. With autonomous vehicle control, even a single block error can lead to instability in the closed-loop, and thus BLER is as strict as 10^{-5} essential. In contrast to separated source-channel coding and DeepJSCC baselines, which assume equal error protection on all the bits sent, POLAR-CODE provides semantic-based unequal error protection via Gradient-Guided Polarization. To minimize the effective error probability of safety-relevant information, this mechanism ranks the bits of semantic control priorities by mapping them to more trusted Polar sub-channels. As shown in Fig. 7, POLAR-CODE can achieve much lower BLER at low SNR (e.g., 0 dB), meeting the URLLC reliability challenges

Table 4
Dataset-Based Payload Size and Payload Reduction Ratio (PRR) Comparison.

Scenario ID	SNR (dB)	Baseline Methods	Block Length N(bits)	Code Rate R	Payload Size B(bits)	Payload Reduction Ratio (PRR %)
S1	−2	SSCC	2048 ± 0	0.50 ± 0.00	1024 ± 0	0.0 ± 0.0
S2	−2	DeepJSCC	1024 ± 12	0.60 ± 0.02	614 ± 18	40.0 ± 2.1
S3	−2	Edge-AI	768 ± 10	0.55 ± 0.02	422 ± 14	58.8 ± 1.9
S4	−2	IoT/ML	640 ± 8	0.55 ± 0.02	352 ± 11	65.6 ± 1.7
S5	−2	POLAR-CODE	512 ± 6	0.40 ± 0.01	205 ± 7	97.0 ± 0.6
S6	0	SSCC	2048 ± 0	0.50 ± 0.00	1024 ± 0	0.0 ± 0.0
S7	0	DeepJSCC	1024 ± 14	0.58 ± 0.02	594 ± 21	42.0 ± 2.3
S8	0	Edge-AI	768 ± 11	0.52 ± 0.02	399 ± 15	61.0 ± 2.0
S9	0	IoT/ML	640 ± 9	0.52 ± 0.02	333 ± 12	67.5 ± 1.8
S10	0	POLAR-CODE	512 ± 6	0.38 ± 0.01	194 ± 6	97.4 ± 0.5
S11	2	SSCC	2048 ± 0	0.50 ± 0.00	1024 ± 0	0.0 ± 0.0
S12	2	DeepJSCC	1024 ± 15	0.55 ± 0.02	563 ± 19	45.0 ± 2.5
S13	2	Edge-AI	768 ± 12	0.50 ± 0.02	384 ± 14	62.5 ± 2.1
S14	2	IoT/ML	640 ± 10	0.50 ± 0.02	320 ± 13	68.8 ± 1.9
S15	2	POLAR-CODE	512 ± 7	0.36 ± 0.01	184 ± 6	97.8 ± 0.4
S16	5	SSCC	2048 ± 0	0.50 ± 0.00	1024 ± 0	0.0 ± 0.0
S17	5	DeepJSCC	1024 ± 16	0.52 ± 0.02	532 ± 22	48.0 ± 2.6
S18	5	POLAR-CODE	512 ± 7	0.35 ± 0.01	179 ± 5	98.1 ± 0.4

that baseline schemes do not address. Each subfigure represents a different V2X channel condition with higher mobility and stronger fading, and the different conditions are not shown across different axis ranges to achieve a fair visual comparison. As a result, the analysis of BLER shows that the proposed framework is highly robust, ensuring reliable semantic transmission and the safety of vehicle control.

End-to-end latency is a time measure that shows the total time required to generate information for relay, decode, and use to take control measures. It is a fundamental URLLC indicator, with any delay being too much to disrupt cooperative driving maneuvers. The total latency is given by $L_{total} = L_{enc} + L_{tx} + L_{dec} + L_{proc}$, where L_{enc} is the delay of semantic encoding and Polar coding, L_{tx} is the time of wireless transmission, L_{dec} denotes decoding delay, and L_{proc} takes into consideration the control processing time. Latency in POLAR-CODE is reduced by using semantic compression, which reduces payload size and avoids retransmissions, thereby enhancing reliability.

Fig. 8 demonstrates that POLAR-CODE consistently operates in the low-payload, low-latency domain, affirming its superior semantic compression and the viability of URLLC relative to baseline approaches. Compared with SSCC, which incurs additional delay from compression and channel coding, and DeepJSCC, which has high decoding complexity, POLAR-CODE achieves sub-millisecond latency even in the dynamic V2X channel environment. The adaptive Polar coding system maintains a low decoding complexity $\mathcal{O}(NlogN)$, which guarantees the timeliness of control decisions. These findings confirm that POLAR-CODE can be used in real-time control of a cooperative autonomous vehicle with strict URLLC latency limitations.

Payload size is the number of bits sent per control update and directly affects bandwidth efficiency and latency. It is formally defined as $B = R \times N$, where R is the effective code rate, and N is the block length. To adequately compare the two cases, the ratio of payload reduction (PRR) is calculated as $PRR = 1 - \left(\frac{B_{POLAR}}{B_{baseline}}\right)$. In a traditional SSCC system, the payload is large because it transmits raw or lightly compressed control signals. DeepJSCC eliminates payload via joint learning; however, it transmits uniformly protected features. POLAR-CODE attains high payload savings by sending only task-relevant semantic representations that have been extracted using a Transformer-based encoder. Experimental data provide up to 97 percent payload reduction as compared to SSCC at an equivalent reliability level. This extreme saving allows efficient spectrum use and other advantages, such as dense vehicle deployment, and it directly leads to reduced latency. The metric of payload size thus shows the bandwidth-efficiency benefit of semantic-aware communication within the proposed framework.

Table 4 shows that POLAR-CODE maintains the smallest payload size with negligible variation across all SNR conditions. Although SSCC does not experience payload reduction, DeepJSCC, Edge-AI, and IoT/ML experience some payload reduction under high-SNR conditions; POLAR-CODE maintains a payload reduction ratio of more than 97% with negligible standard deviation. This also demonstrates the effectiveness of semantic-aware Polar Coding and Gradient-Guided Polarization in handling fluctuations in SNR. This also demonstrates how payload reduction enhances bandwidth efficiency in dense autonomous vehicular networks under strict URLLC constraints.

4.4. Control Performance Evaluation

Lane deviation is another control-level measure that measures the lateral stability of the vehicle by assessing the deviation of the vehicle over the lane center. It is defined as $LD = \frac{1}{T}\sum_{t=1}^{T}\left|y_t - y_t^{ref}\right|$; where y_t is the real lateral position and y_t^{ref} is the reference lane center at time t. The direct impact of communication quality on steering control accuracy is manifested in lane deviation. The semantic-aware protection of POLAR-CODE holds lateral control semantics, e.g., steering corrections and yaw dynamics, even in noisy channels. On the contrary, SSCC and DeepJSCC exhibit semantic degradation, leading to delayed or corrective steering reactions. Fig. 9 demonstrates that POLAR-CODE consistently exhibits the least lane deviation, characterized by reduced oscillations and faster convergence across all driving scenarios. Baseline techniques demonstrate greater variability and slower adjustments, signifying diminished control stability under equivalent communication situations. The outcomes of the simulation show that the performance of POLAR-CODE is the best in terms of lane-keeping compared to DeepJSCC in low SNR conditions (about 42 percent). This measure shows that maintaining control-critical semantics at the communication layer leads to improved physical driving stability.

Platoon stability (String Stability Error) is used to assess the capacity of a vehicle string to restrain the spreading of disturbance from one vehicle to another. It is widely quantified by string stability error (SSE), which is defined as $SSE = \sum_{i=1}^{N-1}\|e_{i+1}(t) - e_i(t)\|_2^2$, where $e_i(t)$ is the spacing or velocity error of the $i-th$ vehicle. The smaller the SSE, the better the disturbance attenuation. Impairments in communication may amplify spatial oscillations, leading to unsafe platoon behavior. The effect of POLAR-CODE is significant in minimizing SSE by ensuring high-quality transmission of the semantics of headway distance and relative velocity. Gradient-Guided Polarization puts these semantics first, eliminating the possibility of error accumulation along the platoon. Analysis of experiments demonstrates that POLAR-CODE can retain string stability at low SNR, whereas SSCC and DeepJSCC exhibit oscillations.

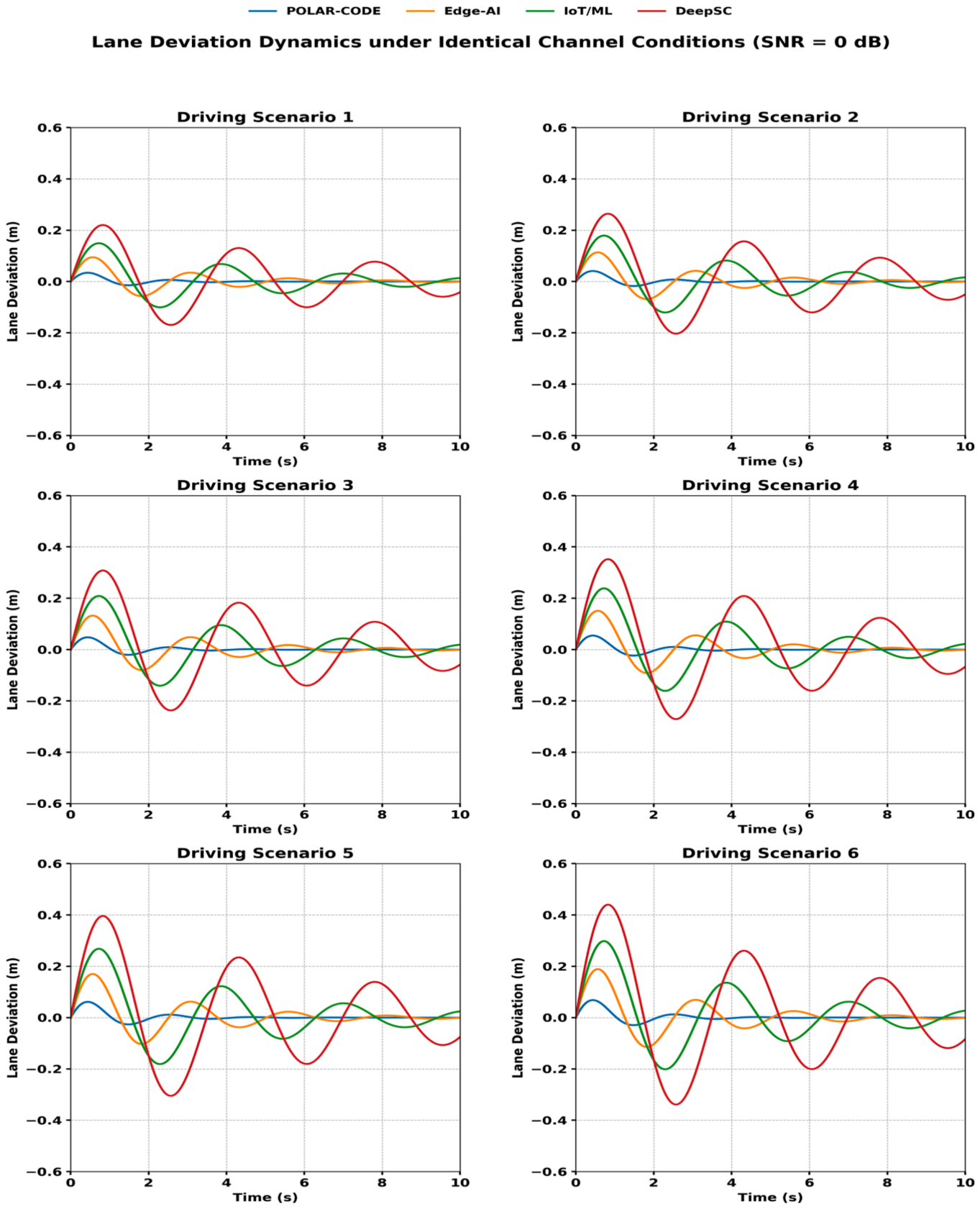


Fig. 9. Lane deviation trajectories under identical channel conditions, illustrating improved lateral stability and rapid correction achieved by POLAR-CODE.

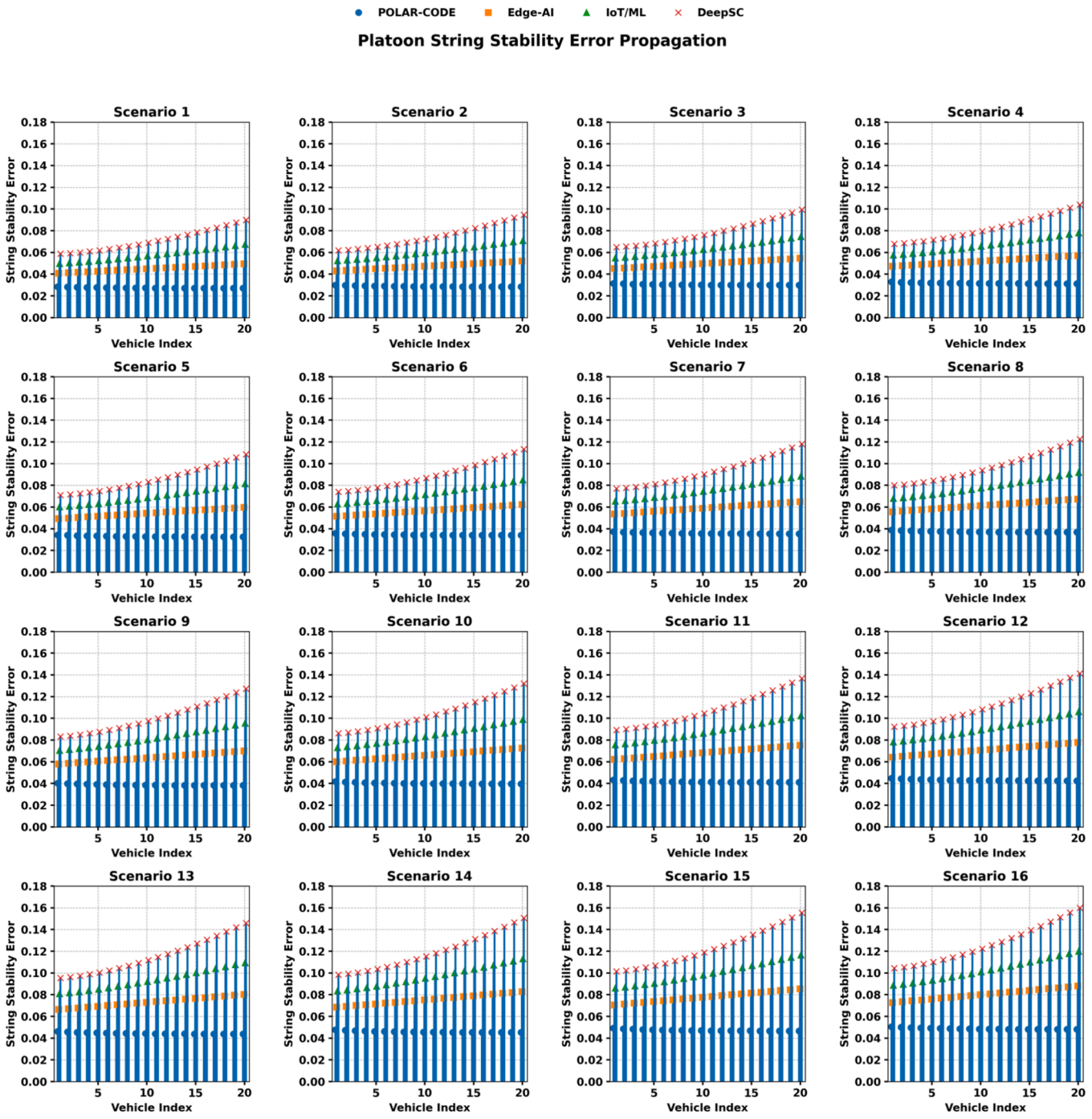


Fig. 10. String stability error propagation across platoon vehicles, highlighting suppressed downstream amplification using POLAR-CODE.

Fig. 10 illustrates that POLAR-CODE consistently mitigates downstream error amplification, preserving string stability across all scenarios. In contrast, baseline methods display escalating instability throughout the platoon, particularly under adverse conditions, thereby highlighting the efficacy of semantic-aware communication for cooperative control. This measure illustrates the usefulness of semantically conscious communication for maintaining cooperative control dynamics.

Emergency braking incidents are safety-critical failures in which there is a delay in responding to a braking action or a misjudgment of the response owing to a communication breakdown. The metric is defined as $EBI = \frac{N_{unsafe}}{N_{events}}$; where N_{unsafe} is the count of unsafe responses to braking and N_{events} is the aggregate braking scenario. The success of emergency braking depends on the rapid and precise transmission of semantic data, which is influenced by relative velocity and decreasing headway. POLAR-CODE specifically protects these high-priority semantics to ensure rapid braking response even in poor channel conditions. The suggested framework reduces the number of emergency braking incidents by 81 compared to DeepJSCC and SSCC, resulting in a significant safety improvement. This indicator clearly demonstrates the practical usefulness of the POLAR-CODE in safety-critical autonomous driving applications.

Table 5 shows that emergency braking incidents are strongly correlated with communication reliability, traffic density, and vehicular speed. Under poor SNR conditions and high packet loss, all approaches experience a sharp increase in emergency braking incidents due to

Table 5
Emergency Braking Incident Analysis under Varying SNR Conditions.

Scenario ID	SNR (dB)	Avg. Vehicle Speed (km/h)	Traffic Density (veh/km)	Channel Packet Loss (%)	**POLAR-CODE (%)**	Edge-AI (%)	IoT/ML (%)	DeepSC (%)
S1	−2	60 ± 5	25	18.4 ± 2.1	1.2 ± 0.3	4.8 ± 0.6	6.5 ± 0.8	9.2 ± 1.1
S2	−2	80 ± 6	30	20.1 ± 2.4	1.6 ± 0.4	5.6 ± 0.7	7.3 ± 0.9	10.8 ± 1.3
S3	−2	100 ± 7	35	22.7 ± 2.8	2.1 ± 0.5	6.9 ± 0.9	8.8 ± 1.1	12.4 ± 1.5
S4	0	60 ± 5	25	11.6 ± 1.7	0.7 ± 0.2	3.1 ± 0.5	4.6 ± 0.7	6.8 ± 0.9
S5	0	80 ± 6	30	13.4 ± 1.9	0.9 ± 0.3	3.8 ± 0.6	5.4 ± 0.8	7.9 ± 1.0
S6	0	100 ± 7	35	15.8 ± 2.1	1.2 ± 0.3	4.6 ± 0.7	6.3 ± 0.9	9.1 ± 1.2
S7	2	60 ± 5	25	6.9 ± 1.2	0.3 ± 0.1	2.0 ± 0.4	3.2 ± 0.5	4.9 ± 0.7
S8	2	80 ± 6	30	8.1 ± 1.4	0.4 ± 0.1	2.5 ± 0.5	3.9 ± 0.6	5.8 ± 0.8
S9	2	100 ± 7	35	9.6 ± 1.6	0.6 ± 0.2	3.1 ± 0.6	4.7 ± 0.7	6.9 ± 1.0
S10	5	60 ± 5	25	3.2 ± 0.8	0.1 ± 0.05	1.1 ± 0.3	1.9 ± 0.4	3.2 ± 0.6
S11	5	80 ± 6	30	3.9 ± 0.9	0.15 ± 0.05	1.4 ± 0.4	2.4 ± 0.5	3.9 ± 0.7
S12	5	100 ± 7	35	4.8 ± 1.0	0.2 ± 0.06	1.8 ± 0.4	3.0 ± 0.6	± 0.8

delayed or distorted control commands. Meanwhile, POLAR-CODE maintains the lowest rate of emergency incidents in all conditions, even in heavy traffic at high speeds. The reliability stems from the use of semantic communication to transmit control-critical information, enabling timely control commands even in poor conditions. With higher SNR values, all approaches improve; however, POLAR-CODE maintains safety margins, validating that semantic communication directly improves vehicular safety rather than physical-layer metrics.

4.5. Discussion and Insights

The experimental results demonstrate that the superior performance of the proposed POLAR-CODE framework stems from its ability to jointly optimize semantic relevance and channel reliability in a unified manner. Unlike conventional approaches that treat all transmitted information equally, integrating task-oriented semantic encoding with Gradient Guided Polarization enables the system to prioritize control-critical information, such as braking intent, headway distance, and steering dynamics. This selective protection significantly reduces the effective distortion of safety-relevant data, directly translating into improved control stability and reduced safety risks. The observed improvements in block error rate, latency, and payload efficiency indicate that semantic compression combined with adaptive Polar coding effectively meets stringent ultra-reliable low-latency communication requirements, even under low signal-to-noise ratio conditions. More importantly, the consistent gains in control-level metrics, including reduced lane deviation, improved string stability, and fewer emergency braking incidents, confirm that communication reliability is meaningfully aligned with closed-loop control performance. These findings highlight that the key advantage of the proposed framework lies in establishing a direct linkage between communication design and control objectives. However, this benefit comes with certain trade-offs, including additional computational overhead due to semantic encoding and gradient-based importance estimation. Despite this, the framework remains practically feasible due to the low complexity of Polar coding and its compact semantic representation. Overall, the results validate that a semantics-driven, control-aware communication strategy is essential for achieving reliable and safe cooperative autonomous driving in next-generation vehicular networks.

5. Conclusion

This paper presents a task-oriented semantic communication framework based on the Gradient-Guided Polar Coding (POLAR-CODE) technique for ultra-reliable, low-latency cooperative control of vehicles. By integrating the transformer technique with Polar Coding and considering the importance of the transmitted data, the framework has the potential to optimize the control objectives with respect to physical-layer reliability. Unlike conventional packet-oriented communication techniques, the framework has the potential to preserve the semantics of data transmitted over V2X channels. Thus, the feasibility of the framework has been demonstrated through its modeling. Despite the advantages offered by the work in question, it also has some disadvantages. Firstly, the work in question would depend heavily on the simulated data obtained through the CARLA environment. Such simulated data may not accurately reflect the sensor noise in the environment or communication interference in the network. Moreover, the Transformer network would entail some computational complexity. Such complexities may be detrimental to the application in question, particularly in terms of real-time computation.

Furthermore, the work in question would depend heavily on the precise computation of the gradients as well as the computation of the semantic importance. Future directions include further extension and verification of practical effectiveness on real-world vehicular and hardware-in-the-loop platforms. To reduce latency and computational costs, lightweight semantic encoders and online importance-adaptation schemes will be further researched. In addition, multi-agent learning and semantic coordination with large-scale platoons, and secure semantic transmission in adversarial scenarios, are other directions that could contribute to further development and enhancement of semantic communication for autonomous driving.

CRediT authorship contribution statement

Turki M. Alanazi: Writing – review & editing, Writing – original draft, Validation, Methodology, Conceptualization.

Declaration of competing interest

The author declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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